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DEPARTMENT OF DEFENSE INTERFACE STANDARD

INTEROPERABILITY AND PERFORMANCE STANDARDS FOR THE DEFENSE INFORMATION SYSTEM

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. Interoperability of DoD telecommunications systems, and of DoD with non-DoD telecommunications systems, has been and will continue to be a major consideration in developing and adopting standards for military use.

a. Military standards (MIL-STD) in the 188 series (MIL-STD-188-XXX) address specific telecommunications design parameters that have been proven to work. MIL-STDs are to be used in all new or major upgrades of inter-and intra-DoD systems and equipment, and are to interface with commercial non-DoD systems and equipment to ensure interoperability.

b. MIL-STDs in the 187 series (MIL-STD-187-XXX) address evolving telecommunications design parameters and concepts that are subject to change and that have not been adequately proven through the use of empirical test data. MIL-STD-187-XXX standards should be used as planning standards and guides until parameters are proven and included in approved federal, allied, MIL-STD-188-XXX, or DoD-adopted commercial standards.

c. MIL-STDs in the 2045 series (MIL-STD-2045-XXXXX) address DoD Communications Protocol Standards (DCPS). A DCPS may be either a base standard or a functional profile. MIL-STD-2045 standards will include enhancements to commercial standards or include new protocol standards that are entirely unique to DoD. The 2045 series also allows for a category of interim DoD standards. Interim standards are needed because of the usual disparity between immediate DoD needs and the amount of time required in the commercial world to adopt new standards.

3. MIL-STD-187-XXX standards provide uniform guidance for the design of the evolving and future Defense Information System (DIS) at the information-transfer level. Providing this guidance at the concept engineering stage will help minimize ineffective designs and costly interoperability problems at later stages of implementation, as well as ensure use of appropriate advances in technology. Planning standards are developed considering current and future plans for the DIS, commercial systems (national and international), and North Atlantic Treaty Organization (NATO) and other allied military systems. These standards are usually based on or make reference to American National Standards Institute (ANSI) standards, International Telecommunications Union-Telecommunication Standardization Sector (ITU-T) [formerly the

International Telegraph and Telephone Consultative Committee (CCITT)] Recommendations, International Organization for Standardization (ISO) standards, NATO standardization agreements (STANAG), and other MIL-STDs, wherever applicable.

4. This MIL-STD contains the technical standards and design objectives necessary to allow strategic and tactical users to exchange all forms of information digitally, using the DIS. The standards contained herein are common to both tactical and strategic systems, unless otherwise specified. This MIL-STD addresses all interoperability elements specified in the DIS framework, except data-processing standards such as standard programming languages and data elements.

5. The standards in this MIL-STD are based on, or make reference to, corresponding parameters in other MIL-STDs, as well as ANSI standards, ITU-T (CCITT) Recommendations, ISO standards, and NATO STANAGs, wherever applicable. Users of this MIL-STD should be aware that there may be patent rights, copyright claims, or both, by companies or individuals on portions of the MIL-STD. Before incorporating this MIL-STD into systems or equipment, users are advised to contact the appropriate standards organization, such as ANSI, regarding claims or conditions that pertain to the use of an applicable commercial standard. Implementers of this MIL-STD are solely responsible for compensating companies or individuals entitled to any royalties.

6. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

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1. SCOPE

1.1 Purpose. The purpose of this military standard (MIL-STD) is to provide a baseline for planning and designing the evolving Defense Information System Network (DISN), defined in 1.7.

1.2 Applicability. This MIL-STD is to be used in planning, designing, and developing new Defense Information System (DIS) communications systems, and in making major changes to existing systems. This MIL-STD does not necessarily apply to leased commercial facilities, but such facilities should be selected to be compatible with its requirements. This MIL-STD applies to digital communications systems only.

1.3 Objectives. This MIL-STD has five objectives:

a. To achieve interoperability between strategic and tactical digital networks for voice, data, facsimile, record traffic, and video services.

b. To provide performance standards for strategic and tactical system users.

c. To adopt specific subsets of commercial standards, where feasible, to achieve cost-effective interoperability, performance, and interfaces.

d. To provide a framework to change existing standards and prepare new standards.

e. To establish a reference source for use by all organizations involved in developing the DISN and procuring DISN-compatible hardware and software.

1.4 System standards and design objectives. When procurement, engineering, or design activities elect to incorporate this planning standard in their acquisition documents, the parameters and requirements specified in this MIL-STD shall be treated as mandatory system standards if the word *shall* is used. Nonmandatory parameters, requirements, and design objectives are indicated by the word *should* (design objectives, rather than standards, are used when there is a lack of measured and verified data or no consensus on the interpretation of the data). *Will* is used to express a declaration of purpose or intent. For a definition of *system standards* and *design objectives*, see Federal Standard (FED-STD)-1037.

1.5 Standards action areas. This MIL-STD addresses the interoperability, performance, and interface standards that should be met by future Department of Defense (DoD) information systems to provide a wide variety of end-to-end digital subscriber services in a single integrated network. These services include voice telephony, data transmission, facsimile,

record traffic, and video teleconferencing (VTC). This MIL-STD addresses standardization in eight major areas:

a. Subscriber services

b. Interfaces, including protocols and voice algorithms

c. Circuit switching and packet switching

d. Transmission

e. Signaling

f. Information security

g. Network management and system control

h. End-to-end performance requirements

In accordance with MIL-STD-970, the standards are based on American National Standards Institute (ANSI) standards; International Telecommunications Union-Telecommunication Standardization Sector (ITU-T) Recommendations for the Integrated Services Digital Network (ISDN); the International Organization for Standardization (ISO) Open Systems Interconnection (OSI) reference model; and existing MIL-STD-188 and MIL-STD-2045 series standards. This MIL-STD references other existing standards (military, federal, commercial, and international). The intent is to avoid duplication of existing standards, ensure backward interoperability, and provide for orderly transition to forward-looking standards for new systems.

1.6 DIS framework. The standards provided in this MIL-STD are based on the DIS framework (see section 4, Figure 1) described below:

a. The DIS concept provides for an evolutionary integration of existing and future DoD computer and telephone communications systems. The Services and Agencies adopted the DIS framework as a guide for developing this MIL-STD. The DIS framework provides efficient, end-to-end, integrated service for information sources, sinks, and processors. Integrated service provides for voice, message, data, graphics, and imagery information-transfer across a single network interface. By definition, the DIS framework includes all components necessary to achieve interoperability between DoD users.

b. The DIS framework consists of three major sections, demarcated by reference points A and B. Users may access the DIS through subscriber network elements, such as source, sink, or processor terminal equipment. These terminal equipment include telephones, facsimile machines, VTC, and other data terminal equipment (DTE). For the information source, sink, or processor

elements to be interoperable, all seven layers of the ISO OSI reference model must be interoperable.

c. DTEs exchange information through information-transfer utilities, which are comprised of local-network elements, wide-network elements, and their respective interoperability reference points. The military Services provide fixed-plant, local-network elements to support strategic users and base operations. They also provide tactical local-network elements to support garrison operations and access to wide-network elements, as well as tactical local-network elements to support deployed combat forces. The Defense Information Systems Agency (DISA) provides wide-network elements to interconnect geographically separated local networks. The wide network includes the Defense Information System Network (DISN) and public switched telephone networks (PSTN). Since the local- and wide-network elements and interoperability reference points in the information-transfer utilities represent the telecommunications portion of DIS, their functionality is limited to the lower three layers of the OSI reference model.

d. Advances in computer and telephone communications technology allow multiple services to be provided by a single network, as in ISDN. Wherever applicable, the DIS framework allows the adoption of ANSI standards for ISDN. Within the DIS framework, circuit-switched voice and data services are based on MIL-STDs for tactical systems and ISDN commercial standards for strategic systems.

1.7 Defense Information Infrastructure (DII)/Defense Information System Network (DISN). The DII is defined as the worldwide aggregation of all mobile and fixed DoD information systems, including sensors, data entry devices, and management and control facilities. The DII allows DoD to collect, produce, store, disseminate, display, and secure information. It consists of communications and all supporting resources, such as network management, control, and value-added services. The DII and DIS are the same, and include the information transport segment (DISN) as well as the information processing segment [such as Corporate Information Management (CIM) megacenters and user terminals] of the information infrastructure.

The DISN includes all telecommunications resources, as described in 4.1. It extends from the wide-area network to the information-transfer and transform functions of the end-user computing and terminal equipment. This equipment includes software and hardware required to interconnect user terminal equipment and software, as well as local-area networks, local-access switches, and radio terminals.

1.8 Waivers for use of specifications and standards. Performance specifications and standards and non-Government standards shall be used instead of detailed Government

specifications and standards. In those cases in which a detailed Government specification or standard (such as military specifications and standards, program-unique detailed specifications and standards, and federal specifications and standards) is needed in a solicitation to define an exact design solution, the DoD program office or buying command may use a Government specification or standard, but only if a waiver is granted. The Standards Improvement Executives (SIE) for the military departments and Defense Logistics Agency may exempt a specification or standard from the waiver process for their use in the solicitation, as they deem necessary. DoD components that do not have an SIE may request a similar exemption from the chairman of the Defense Standards Improvement Council. These exemptions must be renewed at least every two years. The following types of documents do not require waivers:

a. Any document required by law, the Federal Acquisition Regulation, or the Defense Federal Acquisition Regulation Supplement.

b. Non-Government standards.

c. Federal Information Processing Standards.

d. Government specifications designated in the DoD Index of Specifications and Standards (DoDISS) as performance specifications.

e. Commercial item descriptions listed in the DoDISS.

f. Guide specifications listed in the DoDISS.

g. Interface standards listed in the DoDISS.

h. Standard Practices listed in the DoDISS.

i. Guides listed in the DoDISS.

j. Handbooks listed in the DoDISS.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this standard, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto, cited in the solicitation.

STANDARDS

FEDERAL

FED-STD-1002	Time and Frequency Reference Information in Telecommunication Systems
FED-STD-1016	Telecommunications: Analog to Digital Conversion of Radio Voice by 4,800 Bit/Second Code Excited Linear Prediction (CELP)
FED-STD-1037	Glossary of Telecommunication Terms
FED-STD-1047	Telecommunications: HF Radio Automatic Message Exchange (Draft)
FED-STD-1048	Telecommunications: HF Radio Automatic Networking to Multimedia (Draft)
FED-STD-1055	Telecommunications: Interoperability Requirements for Meteor Burst Radio Communications Between Conventional Master and Remote Stations
FED-STD-1056	Telecommunications: Interoperability Requirements for the Encryption of Meteor Burst Radio Communications

FED-STD-1057	Telecommunications: Interoperability Requirements for Meteor Burst Radio Communications Between Networks by Master Stations

FEDERAL INFORMATION PROCESSING STANDARDS PUBLICATIONS (FIPS PUBS)

FIPS PUB 146	Profile for Open Systems Internetworking Technologies (POSIT)
FIPS PUB 179	Government Network Management Profile (GNMP)
FIPS PUB 182	Integrated Services Digital Network (ISDN)

DEPARTMENT OF DEFENSE

MIL-STD-187-721	Planning and Guidance Standard for Automated Control Applique for HF
MIL-STD-188-105	All-Digital Tactical-to-Strategic Gateway
MIL-STD-188-110	Interoperability and Performance Standards for Data Modems
MIL-STD-188-111*	Interoperability and Performance Standards for Fiber Optic Communications Systems
MIL-STD-188-112	Subsystem Design and Engineering Standards for Common Long Haul/Tactical Cable and Wire Communications
MIL-STD-188-113	Interoperability and Performance Standards for Analog-to-Digital Conversion Techniques
MIL-STD-188-114	Electrical Characteristics of Digital Interface Circuits
MIL-STD-188-115	Interoperability and Performance Standards for Communications Timing and Synchronization Subsystems

* NOTE: This standard is not listed in the DoDISS as an interface standard; thus, it requires a waiver if used in a solicitation.

MIL-STD-188-124*	Grounding, Bonding and Shielding for Common Long Haul/Tactical Communication Systems Including Ground Based Communications-Electronics Facilities and Equipments
MIL-STD-188-136	EHF Medium Data Rate (MDR) Satellite Data Link Standards (SDLS): Uplinks and Downlinks (Draft)
MIL-STD-188-140	Equipment Technical Design Standards for Common Long Haul/Tactical Radio Communications in the Low Frequency Band and Lower Frequency Bands
MIL-STD-188-141	Interoperability and Performance Standards for Medium and High Frequency Radio Equipment
MIL-STD-188-145	Interoperability and Performance Standards for Digital LOS Microwave Radio Equipment
MIL-STD-188-148	Interoperability Standard for AJ Communications in the High Frequency (2-30 MHz) Band (U), SECRET
MIL-STD-188-161	Interoperability and Performance Standards for Digital Facsimile Equipment
MIL-STD-188-164	Interoperability and Performance Standards for C-Band, X-Band, and Ku-Band SHF Satellite Communications Earth Terminals
MIL-STD-188-165	Interoperability and Performance Standards for SHF Satellite Communications PSK Modems (FDMA Operation)
MIL-STD-188-166	Interoperability and Performance Standards for SHF SATCOM Link Control (Draft)

* NOTE: This standard is not listed in the DoDISS as an interface standard; thus, it requires a waiver if used in a solicitation.

MIL-STD-188-167	Interoperability and Performance Standards for SHF SATCOM Demand Assignment (Draft)
MIL-STD-188-168	Interoperability and Performance Standards for SHF SATCOM Multiplexer (Draft)
MIL-STD-188-169	Interoperability and Performance Standards for C-Band and Ku-Band Satellite Communications Nonprocessing Transponders (Draft)
MIL-STD-188-171	Interoperability Standards for Information and Record Traffic Exchange, Mode I
MIL-STD-188-172	Interoperability Standards for Information and Record Traffic Exchange, Mode II
MIL-STD-188-173	Interoperability Standards for Information and Record Traffic Exchange, Mode V
MIL-STD-188-174	Interoperability Standards for Information and Record Traffic Exchange, Mode VI
MIL-STD-188-181	Interoperability Standard for Dedicated 5-kHz and 25-kHz UHF Satellite Communications Channels
MIL-STD-188-182	Interoperability Standard for 5-kHz UHF DAMA Terminal Waveform
MIL-STD-188-183	Interoperability Standard for 25-kHz UHF TDMA/DAMA Terminal Waveform
MIL-STD-188-184	Interoperability and Performance Standard for the Data Control Waveform
MIL-STD-188-185	Interface Interoperability Standard for UHF MILSATCOM DAMA Control System (Draft)
MIL-STD-188-190*	Methods for Communications Systems Measurements

* NOTE: This standard is not listed in the DoDISS as an interface standard; thus, it requires a waiver if used in a solicitation.

MIL-STD-188-196	NITFS, Bi-Level Image Compression
MIL-STD-188-197	NITFS, Adaptive Recursive Interpolated Differential Pulse Code Modulation (ARIDPCM)
MIL-STD-188-198	NITFS, Joint Photographic Experts Group (JPEG) Image Compression
MIL-STD-188-199	NITFS, Vector Quantization Decompression
MIL-STD-188-200	System Design and Engineering Standards for Tactical Communications
MIL-STD-188-202	Interoperability and Performance Standards for Tactical Digital Transmission Groups (Coaxial Cable)
MIL-STD-188-203-1	Interoperability and Performance Standards for Tactical Digital Information Link (TADIL) A
MIL-STD-188-203-3	Subsystem Design and Engineering Standards for Tactical Digital Information Link (TADIL) C
MIL-STD-188-212	Subsystem Design and Engineering Standards for Tactical Digital Information Link (TADIL) B
MIL-STD-188-216	Interoperability Standards for Data Adapter Control Mode
MIL-STD-188-220	Interoperability Standard for Digital Message Transfer Device Subsystems
MIL-STD-188-242	Interoperability and Performance Standards for Tactical Single Channel Very High Frequency (VHF) Radio Equipment
MIL-STD-188-243	Interoperability and Performance Standards for Tactical Single Channel Ultra High Frequency (UHF) Radio Communications
MIL-STD-188-256	Interoperability and Performance Standards for Digital Signaling and Supervision of Tactical Communications Systems

MIL-STD-188-260	Design and Engineering Standards for Tactical Terminal Subsystems
MIL-STD-188-311	Technical Design Standards for Frequency-Division Multiplexers
MIL-STD-210	Climatic Information to Determine Design and Test Requirements for Military Systems and Equipment
MIL-STD-449*	Radio Frequency Spectrum Characteristics, Measurement of
MIL-STD-461*	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462*	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781*	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-785*	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810*	Environmental Test Methods and Engineering Guidelines
MIL-STD-1472*	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1582	EHF Low Data Rate (LDR) Satellite Data Link Standards (SDLS) Uplinks and Downlinks (SECRET)
MIL-STD-2045-13500	Information Technology - DoD Standardized Profiles - Internet Relay Profiles
MIL-STD-2045-13500-1	Subnetwork Independent Requirements for Relaying IP

* NOTE: This standard is not listed in the DoDISS as an interface standard; thus, it requires a waiver if used in a solicitation.

MIL-STD-2045-13500-2	Subnetwork Dependent Requirements for the Point-to-Point Protocol (PPP)
MIL-STD-2045-13501	Information Technology - DoD Standardized Profiles - Internet Routing Between Autonomous Systems
MIL-STD-2045-14500 Series	Transport Profiles
MIL-STD-2045-14500-1	Information Technology - DoD Standardized Profiles TA21(D) Transport Profile Part 1: Reliable End System (ES) Transport
MIL-STD-2045-14500-2	Information Technology - DoD Standardized Profiles TA21(D) Transport Profile Part - COTS Over CLNS Part 2: Balanced Point-to-Point Digital Data Circuit
MIL-STD-2045-14500-3	Information Technology - DoD Standardized Profiles TA21(D) Transport Profile - COTS Over CLNS Part 3: Subnetwork for an Unbalanced Link
MIL-STD-2045-14500-4	Information Technology - DoD Standardized Profiles TA21(D) Transport Profile Part 4: Local Area Networks (LANs), Using Token Bus
MIL-STD-2045-14500-5	Information Technology - DoD Standardized Profiles TA21(D) Transport Profile Part 5: Local Area Networks (LANs), Using Token Ring
MIL-STD-2045-14500-6	Information Technology - DoD Standardized Profiles TA21(D) Transport Profile Part 6: Integrated Services Digital Network (ISDN)
MIL-STD-2045-14502 Series	Internet Transport Profiles

MIL-STD-2045-14502-1	Information Technology - DoD Standardized Profiles - Internet Transport Profile for DoD Communications - Part 1: Transport and Internet Services
MIL-STD-2045-14502-2	Information Technology - DoD Standardized Profiles - Internet Transport Profile for DoD Communications - Part 2: Point-to-Point Links
MIL-STD-2045-14502-3	Information Technology - DoD Standardized Profile - Internet Transport Profile - Part 3: Wide Area Network Access
MIL-STD-2045-14502-4	Information Technology - DoD Standardized Profile - Internet Transport Profile - Part 4: IEEE 802 Local Area Network (LAN) Media Independent Requirements
MIL-STD-2045-14502-5	Information Technology - DoD Standardized Transport Profile - Internet Transport Profile for DoD Communications - Part 5: IEEE 802 Local Area Network (LAN) Media Dependent Requirements
MIL-STD-2045-14502-6	Information Technology - DoD Standardized Profile - Internet Transport Profile for DoD Communications - Part 6: Combat Net Radio (CNR)
MIL-STD-2045-14503	Information Technology - DoD Standardized Profiles - Internet Transport Service Supporting OSI Applications
MIL-STD-2045-17503 Series	Internet Message Transport Profile
MIL-STD-2045-17503-1	Information Technology - DoD Standardized Profiles - Internet Message Transfer Profile for DoD Communications - Part 1: Simple Mail Transfer Protocol (SMTP)

MIL-STD-2045-17503-2	Information Technology - DoD Standardized Profiles - Internet Message Transfer Profile for DoD Communications - Part 2: Format of Text Messages
MIL-STD-2045-17504	Information Technology - DoD Standardized Profiles - Internet File Transfer Profile for DoD Communications
MIL-STD-2045-17505	Information Technology - DoD Standardized Profiles - Internet Domain Name Service (DNS) Profile for DoD Communications
MIL-STD-2045-17506	Information Technology - DoD Standardized Profile - Internet Remote Login Profile for DoD Communications
MIL-STD-2045-17507 Series	Internet Network Management Profiles
MIL-STD-2045-17507-1	Information Technology - DoD Standardized Profiles - Internet Network Management Profile for DoD Communications - Part 1: Simple Network Management Protocol (SNMP)
MIL-STD-2045-17507-2	Information Technology - DoD Standardized Profiles - Internet Network Management Profile for DoD Communications - Part 2: Management Information Base (MIB)
MIL-STD-2045-17507-3	Information Technology - DoD Standardized Profiles - Internet Network Management Profile for DoD Communications - Part 3: Structure and Identification of Management Information
MIL-STD-2045-17508	Information Transfer DoD Profiles AFT1n(D) - File Transfer, Access and Management (FTAM)

MIL-STD-2045-17508-1	Information Technology - DoD Standardized Profiles AFT1n(D) - File Transfer, Access and Management - Part 1: Specification of ACSE, Presentation and Session Protocols for use by FTAM
MIL-STD-2045-17508-2	Information Technology - DoD Standardized Profiles AFT1n(D) - File Transfer, Access and Management - Part 2: Definition of Document Types, Constraint Sets and Syntaxes
MIL-STD-2045-17508-3	Information Technology - DoD Standardized Profiles AFT1n(D) - File Transfer, Access and Management - Part 3: AFT11 - Simple File Transfer Service (unstructured)
MIL-STD-2045-17508-4	Information Technology - DoD Standardized Profiles AFT1n(D) - File Transfer, Access and Management - Part 4: Positional File Transfer Service for Flat Files
MIL-STD-2045-17508-5	Information Technology - DoD Standardized Profiles AFT1n(D) - File Transfer, Access and Management - Part 5: Positional File Access Service for Flat Files
MIL-STD-2045-17508-6	Information Technology - DoD Standardized Profiles AFT1n - File Transfer, Access and Management - Part 6: AFT3 - File Management Service
MIL-STD-2045-18500 Series	Message Handling Systems
MIL-STD-2045-18500-1	Information Technology DoD Standardized Profiles AMHXn(D) - Message Handling Systems - Message Security Protocol - Part 1: MSP Service Support

MIL-STD-2045-18500-2	Information Technology DoD Standardized Profiles AMHXn(D) - Message Handling Systems - Message Security Protocol - Part 2: AMHX1(D) - MSP Content Protocol
MIL-STD-2045-18500-3	Information Technology DoD Standardized Profiles AMHXn(D) - Message Handling Systems - Message Security Protocol - Part 3: AMHX2(D) - MSP Requirements for Message Transfer (P1)
MIL-STD-2045-18500-4	Information Technology DoD Standardized Profiles AMHXn(D) - Message Handling Systems - Message Security Protocol - Part 4: AMHX3(D) - MSP Requirements for MTS Access (P3)
MIL-STD-2045-18500-5	Information Technology DoD Standardized Profiles AMHXn(D) - Message Handling Systems - Message Security Protocol - Part 5: AMHX4(D) - MSP Requirements for MS Access (P7)
MIL-STD-2045-38000	DoD Network Management for DoD Communications
MIL-STD-2045-44500	Tactical Communications Protocol 2 (TAC02) for the NITFS
MIL-STD-2045-47001	Connectionless Data Transfer, Application Layer Standard
MIL-STD-2045-48501	Common Security Label
MIL-STD-2301	Computer Graphics Metafile Implementation for the National Imagery Transmission Format Standard
MIL-STD-2500	National Imagery Transmission Format Standard
MIL-STD-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities

HANDBOOKS

MIL-HDBK-232	RED/BLACK Engineering-Installation Guidelines
MIL-HDBK-235	Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems and Systems
MIL-HDBK-237	Electromagnetic Compatibility Management Guide for Platforms, Systems and Equipments
MIL-HDBK-241	Design Guide for Electromagnetic Interference (EMI) Reduction in Power Supplies
MIL-HDBK-253	Guidance for the Design and Test of Systems Protected Against the Effects of Electromagnetic Energy
MIL-HDBK-419	Vol I, Grounding, Bonding and Shielding for Electronic Equipments and Facilities Basic Theory; and Vol II, Grounding Bonding, and Shielding for Electronic Equipments and Facilities Applications
MIL-HDBK-470*	Maintainability Program for Systems and Equipment
MIL-HDBK-471*	Maintainability Verification/ Demonstration/Evaluation
MIL-HDBK-829A	Guidelines for Developing Data Communications Protocol Standards
MIL-HDBK-1300	National Imagery Transmission Format Standard (NITFS)
MIL-HDBK-1350-1	Validation of Data Communications Protocol Standards for Military Applications
MIL-HDBK-1350-2	Data Communications Protocol Conformance and Interoperability Testing and Registration, Volume 2, September 1993 (Draft)

* NOTE: This standard is not listed in the DoDISS as an interface standard; thus, it requires a waiver if used in a solicitation.

MIL-HDBK-1351	Network Management for DoD Communications

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

(Copies of the Federal Information Processing Standards (FIPS) are available to Department of Defense activities from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Others must request copies from the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161-2171.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CJCSM 6231	Manual for Employing Joint Tactical Communications
DCAC 370-175-13	Defense Switched Network (DSN) System Interface Criteria
DISN Architecture	Defense Information System Network (DISN) Architecture, DISN-AR-1000, 12 May 1993 (Draft)
DoD 5200.28-STD	Department of Defense Trusted Computer System Evaluation Criterion
TAFIM	Technical Architecture Framework for Information Management (TAFIM), Volume 7, Adopted Information Technology Standards (AITS), Version 2.0, 12 November 1994
JIEO Specification 9001	Joint Technical Interface Specification for VHF SINCGARS Waveform
JIEO Specification 9109	Technical Interface Specification: Joint Interoperability via Fiber Optic Cable
PG/6 TCP 2000	Tri-Service Group on Communications and Electronics, Project Group 6 -- Post-2000

(To obtain other DoD publications not found in the DoDISS, contact the Defense Information Systems Agency, Center for Standards, ATTN: JEBBB, Fort Monmouth, NJ 07703-5613.)

NACSEM 5201	TEMPEST Guidelines for Equipment/ System Design (U)
NSTISSAM TEMPEST/ 1-91	Compromising Emanations Laboratory Test Requirements, Electromagnetics (Classified document)

(Requests for NACSEM 5201 and NSTISSAM TEMPEST/1-91 should be submitted to the National TEMPEST Information Center, Attention: C941, National Security Agency, Fort George Meade, MD 20899.)

2.2.3 Standardization Agreements (STANAG) and Allied Communication Publications (ACP). Currently all these STANAGs are not listed in the DoDISS but will be incorporated as soon as possible.

STANAGs

STANAG 4175	Technical Characteristics of the Multi-Functional Information Distribution System (MIDS)
STANAG 4206	The NATO Multi-Channel Tactical Digital Gateway System Standards
STANAG 4207	The NATO Multi-Channel Tactical Digital Gateway Multiplex Group Framing Standards
STANAG 4208	The NATO Multi-Channel Tactical Digital Gateway Signalling Standards
STANAG 4209	The NATO Multi-Channel Tactical Digital Gateway Standards for Analogue-to-Digital Conversion of Speech Signals
STANAG 4210	The NATO Multi-Channel Tactical Digital Gateway Cable Link Standards
STANAG 4211	The NATO Multi-Channel Tactical Digital Gateway System Control Standards

...ation number used individually or collectively to ... derived its patent in modern practice the ...

STANAG 4212	The NATO Multi-Channel Tactical Digital Gateway Radio Relay Link Standards
STANAG 4213	The NATO Multi-Channel Tactical Digital Gateway Data Transmission Standards
STANAG 4214	International Routing and Directory for Tactical Communications Systems
STANAG 4249	The NATO Multi-Channel Tactical Digital Gateway -- Data Transmission Standards (Packet Switching Service)
STANAG 4251	NATO Reference Model for Open Systems Interconnection -- Layer 1 (Physical Layer) Service Definition
STANAG 4252	NATO Reference Model for Open Systems Interconnection -- Layer 2 (Data Link Layer) Service Definition
STANAG 4253	NATO Reference Model for Open Systems Interconnection -- Layer 3 (Network Layer) Service Definition
STANAG 4255	NATO Reference Model for Open Systems Interconnection -- Layer 5 (Session Layer) Service Definition (Draft)
STANAG 4256	NATO Reference Model for Open Systems Interconnection -- Layer 6 (Presentation Layer) Service Definition (Draft)
STANAG 4259	NATO Reference Model for Open Systems Interconnection Encoding Rules for ASN.1
STANAG 4261	NATO Reference Model for Open Systems Interconnection -- Layer 1 (Physical Layer) Protocol Specification
STANAG 4262	NATO Reference Model for Open Systems Interconnection -- Layer 2 (Data Link Layer) Protocol Specification; Annex D, Data Link Access Procedure Balanced (LAPB)

STANAG 4263	NATO Reference Model for Open Systems Interconnection -- Layer 3 (Network Layer) Protocol Specification, Annex D, X.75 Packet Level Protocol (STE-STE)
STANAG 4265	NATO Reference Model for Open Systems Interconnection -- Layer 5 (Session Layer) Protocol Specification (Draft)
STANAG 4266	NATO Reference Model for Open Systems Interconnection -- Layer 6 (Presentation Layer) Protocol Specification (Draft)
STANAG 4290	The NATO Multi-Channel Tactical Digital Gateway Cable Link (Optical) Standards (Drafts)
STANAG 4372	Second-generation Anti-jam Tactical UHF Radio for NATO (SATURN)
STANAG 5000	Interoperability of Tactical Digital Facsimile Equipment
STANAG 5516	Tactical Data Exchange Link-16

(Copies of STANAGs, required by contractors in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting officer.)

ACPs

ACP 123 U.S. Supplement No. 1	Common Messaging Strategy and Procedures

(Requests for Allied Communication Publications should be submitted to the DISA Center for Standards, ATTN: JEBBB, Fort Monmouth, NJ 07703-5613.)

2.2.4 NIST publications. (NIST publications require a waiver when used in a solicitation.)

NIST IR90-4250	Network Transport and Message Security Protocols
NIST Special Publication 500-183	National Institute of Standards and Technology (NIST) Special Publication 500-183, Stable Implementation Agreements for Open Systems Interconnection Protocols, Version 4, Edition 1

(NIST documents can be obtained from the National Technical Information Service (NTIS), 5285 Port Royal Road, Springfield, VA 22161-2171 or by calling 1-800-553-6847.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation. (Non-Government standards do not require a waiver when used in a solicitation.)

2.3.1 ITU-T (formerly known as CCITT) Recommendations. The ITU-T is part of the United Nations, a treaty organization. The United States government participates in it through the Department of State, and although industry representatives may work on its committees, approval of standards (called Recommendations) is by governments.

ITU-T D.280	Charging, billing, accounting, and reimbursement for UPT
ITU-T E.163	Numbering Plan for the International Telephone Service
ITU-T E.164	Numbering Plan for the ISDN Era
ITU-T E.168	UPT numbering
ITU-T E.174	Routing principle and guidance for UPT
ITU-T E.751	Reference connections for engineering of land mobile networks (Draft)
ITU-T E.755	Reference connections for UPT
ITU-T E.771	Network grade-of-service parameters and target values for circuit-switched public land mobile services (Draft)
ITU-T E.775	UPT grade-of-service concept
ITU-T E.776	Grade-of-service parameters for networks supporting UPT
ITU-T E.780	Traffic engineering methods for land mobile systems (Draft)

transducers mounted nearly horizontally and from a great distance come to a focus in the

ITU-T E.785	Traffic engineering methods for networks supporting UPT
ITU-T F.69	Plan for Telex Destination Codes
ITU-T F.115	Service objectives and principles for FPLMTS
ITU-T F.724	Videotelephony services for FPLMTS (Draft)
ITU-T F.850	UPT service principle
ITU-T F.851	UPT service set 1
ITU-T F.852	UPT service set 2
ITU-T F.853	UPT supplementary service
ITU-T F.sfea	Service features in FPLMTS (Draft)
ITU-T FPLMTS.TMLG	Vocabulary of terms for FPLMTS (Draft)
ITU-T FPLMTS.FMGM	Framework of FPLMTS management (Draft)
ITU-T FPLMTS.RSEL	Procedure for evaluation of radio transmission technologies for FPLMTS (Draft)
ITU-T FPLMTS.SFMK	Framework for the satellite component of FPLMTS (Draft)
ITU-T FPLMTS.SECMOP	Security mechanisms and operating procedures for FPLMTS
ITU-T G.703	Physical/Electrical Characteristics of Hierarchical Digital Interfaces
ITU-T G.704	Synchronous Frame Structures Used at Primary and Secondary Hierarchical Levels
ITU-T G.707	Synchronous Digital Hierarchy Bit Rates
ITU-T G.708	Network Node Interface for the Synchronous Digital Hierarchy
ITU-T G.709	Synchronous Multiplexing Structure

ITU-T G.711	Pulse Code Modulation (PCM) of voice frequencies
ITU-T G.721	32 kbps Adaptive Pulse Code Modulation
ITU-T G.728	Coding of speech at 16 kbps using low-delay code-excited linear prediction (CELP)
ITU-T G.811	Timing Requirements at the Outputs of Primary Reference Clocks Suitable for Plesiochronous Operation of International Digital Links
ITU-T H.26P/M	Extension of H.263 for mobile application (Draft)
ITU-T I.11x	Vocabulary of terms for mobile networks (Draft)
ITU-T I.37w	Network architecture and capabilities for FPLMTS (Draft)
ITU-T I.5xw	Network interworking between FPLMTS and other types of networks (Draft)
ITU-T I.137	UPT network capabilities
ITU-T I.144	Vocabulary-UPT
ITU-T I.252	Call Offering Supplementary Services
ITU-T I.254	Multiparty Supplementary Services
ITU-T I.321	Broadband Integrated Services Digital Network (B-ISDN) Protocol Reference Model and Its Application
ITU-T I.361	Physical Layer Specification
ITU-T I.363	B-ISDN ATM Adaptation Layer (AAL) Specification
ITU-T I.432	B-ISDN User-Network Interface Physical Layer Specification
ITU-T I.460	Multiplexing, Rate Adaptation and Support of Existing Interfaces
ITU-T M.687-1	Future Public Land Mobile Telecommunication Systems (FPLMTS)

coordinates, either rectangular or polar their significance. Also called

ITU-T M.816	Framework for services supported on FPLMTS
ITU-T M.817	FPLMTS network architectures
ITU-T M.818-1	Satellite operation within FPLMTS
ITU-T M.819-1	FPLMTS for developing countries
ITU-T M.1034	Requirements for the radio interface(s) for FPLMTS
ITU-T M.1035	Framework for the radio interface(s) and radio subsystem functionality for FPLMTS
ITU-T M.1036	Spectrum considerations for implementation of FPLMTS in the bands 1885-2025 MHz and 2110-2200 MHz
ITU-T M.1078	Security principles for FPLMTS
ITU-T M.1079	Speech and voiceband data performance requirements for FPLMTS
ITU-T M.32xx	TMN management service for FPLMTS
ITU-T Q.76	Service procedures for UPT
ITU-T Q.704	Signalling Network Functions and Messages
ITU-T Q.774	Transaction Capabilities Procedure
ITU-T Q.920	ISDN User-Network Interface Data Link Layer -- General Aspects
ITU-T Q.921	ISDN User-Network Interface -- Data Link Layer Specification
ITU-T Q.922	ISDN-Data Link Layer Specification for Frame Mode Bearer Service
ITU-T Q.931	ISDN User-Network Interface Layer 3 Specification for Basic Call Control
ITU-T Q.2130	Service-Specific Coordination Function (SSCF) for Signaling at the UNI

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ITU-T Q.2140	Service-Specific Coordination Function (SSCF) for Signaling at the NNI
ITU-T Q.2761 to Q.2764	BISDN NNI Network Signaling Requirements
ITU-T Q.2931	BISDN UNI Layer 3 Access Call Control Requirements
ITU-T Q.2971	Point-to-Multipoint Call Connection Control (Draft)
ITU-T Q.FIF	FPLMTS information flows (Draft)
ITU-T Q.FNA	FPLMTS network architecture (Draft)
ITU-T Q.UPT	Stage 3 for support of UPT service set 1 on IN CS1
ITU-T V.35	Data Transmission at 48 Kilobits Per Second Using 60-108 kHz Group Band Circuits
ITU-T V.110	Support of Data Terminal Equipments (DTEs) with V-Series Type Interfaces by an Integrated Services Digital Network (ISDN)
ITU-T X.25	Interface Between Data Terminal Equipment (DTE) and Data Circuit-Terminating Equipment (DCE) for Terminals Operating in the Packet Mode and Connected to Public Data Networks by Dedicated Circuit
ITU-T X.31	Support of Packet Mode Terminal Equipment by an ISDN
ITU-T X.75	Packet-Switched Signalling System Between Public Networks Providing Data Transmission Services
ITU-T X.121	International Numbering Plan for Public Data Networks
ITU-T X.217	Association control Service Definition for Open Systems Interconnection for CCITT Applications
ITU-T X.219	Remote Operations: Model, Notation and Service Definition

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ITU-T X.224	Transport Protocol Specification for Open Systems Interconnection for ITU-T Applications
ITU-T X.290	OSI Conformance Testing Methodology and Framework for Protocol Recommendations for ITU-T Applications
ITU-T X.400	Message Handling System and Service Overview
ITU-T X.435	Electronic Data Interchange (EDI) (Draft)
ITU-T X.500	The Directory -- Overview of Concepts, Models and Services

(Copies of International Telegraph and Telephone Consultative Committee (CCITT) standards may be obtained from the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.) [Note: The CCITT has changed its name to the International Telecommunications Union - Telecommunication Standardization Sector (ITU-T).]

2.3.2 American National Standards Institute (ANSI) standards

ANSI T1.101	Synchronization Interface Standards for Digital Service
ANSI T1.105	Digital Hierarchy -- Optical Interface Rates and Formats Specifications
ANSI T1.105.01	Synchronous Optical Network (SONET) -- Automatic Protection Switching
ANSI T1.105.03	Synchronous Optical Network (SONET) -- Jitter at Network Interfaces
ANSI T1.105.09	Synchronous Optical Network (SONET) -- Network Element Timing and Synchronization (Draft)
ANSI T1.106	Digital Hierarchy -- Optical Interface Specifications (Single Mode)
ANSI T1.107	Digital Hierarchy Format Specifications

ANSI T1.111	Signalling System Number 7 (SS7) -- Message Transfer Part (MTP)
ANSI T1.112	Signalling System Number 7 (SS7) -- Signalling Connection Control Part (SCCP)
ANSI T1.113	Signalling System Number 7 (SS7) -- Integrated Services Digital Network (ISDN) User Part
ANSI T1.114	Signalling System Number 7 (SS7) -- Transaction Capability Application Part (TCAP)
ANSI T1.408	ISDN Primary Rate -- Customer Installation Metallic Interfaces, Layer 1 Specification
ANSI T1.601	Integrated Services Digital Network (ISDN) -- Basic Access Interface for Use on Metallic Loops for Application on the Network Side of the NT (Layer 1 Specification)
ANSI T1.602	Integrated Services Digital Network (ISDN) -- Data-Link Layer Signalling Specification for Application at the User-Network Interface
ANSI T1.605	Integrated Services Digital Network (ISDN) -- Basic Access Interface for S and T Reference Points (Layer 1 Specification)
ANSI T1.606	Integrated Services Digital Network (ISDN) -- Architectural Framework and Service Description for Frame-Relaying Bearer Service
ANSI T1.607	Digital Subscriber Signalling System No. 1 -- Layer 3 Signalling Specification for Circuit Switched Bearer Service
ANSI T1.608	Digital Subscriber Signalling System No. 1 (DSS1) -- Signalling Specification for X.25 Packet Switched Bearer Service

ANSI T1.609	Interworking Between the ISDN User -- Network Interface Protocol and the Signalling System Number 7 ISDN User Part
ANSI T1.610	Digital Subscriber Signalling System No. 1 (DSS1) -- Generic Procedures for the Control of ISDN Supplementary Services
ANSI T1.613	Call Waiting Supplementary Service
ANSI T1.616	Call Holding Supplementary Service
ANSI T1.617	Integrated Services Digital Network (ISDN) -- Digital Subscriber Signaling System No. 1 (DSS1) -- Signaling Specification for Frame Relay Bearer Service
ANSI T1.618	Integrated Services Digital Network (ISDN) -- Core Aspects of Frame Protocol for Use with Frame Relay Bearer Service
ANSI T1.619	Integrated Services Digital Network (ISDN) -- Multi-Level Precedence and Preemption (MLPP) Service Capability
ANSI T1.621	Integrated Services Digital Network (ISDN) -- User-to-User Supplementary Service
ANSI T1.622	Integrated Services Digital Network (ISDN) -- Message Waiting Indicator Control and Notification Supplementary Service and Associate Switching and Signaling Specification
ANSI T1.625	Integrated Services Digital Network (ISDN) -- Calling Line Presentation and Restriction Supplementary Services
ANSI T1.627	BISDN - ATM Layer Functionality and Specification
ANSI T1.629	BISDN -- ATM Adaptation Layer 3/4 Common Part Functions and Specifications

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ANSI T1.630	BISDN -- ATM Adaption Layer for Constant Bit Rate Services Functionality and Specification
ANSI T1.632-93	ISDN Supplementary Service Normal Call Transfer
ANSI T1.633-93	Frame Relaying Bearer Service Interworking (I.555)
ANSI T1.634-93	Frame Relaying Service-Specific Convergence Sublayer (FR-SSCS) (I.365.1)
ANSI T1.635-94	B-ISDN - AAL-5 Common Part Functions and Spec (I.363, Section 6)
ANSI T1.637-94	B-ISDN Service-Specific Connection-Oriented Protocol (SSCOP) Spec
ANSI T1.642-94	Call Deflection Supplementary Services
ANSI T1S1.2/91-408	Hotline Service Integrated Text
ANSI T1S1.1/92-188	Proposed Integrated Text for Line Hunting Service with Issue Section
ANSI T1S1.1/92-253	Call Completion to Busy Subscriber, Stage 1/2
ANSI T1S1.2/92-323	Call Completion to Busy Subscriber, DSS1, Stage 3 Description
ANSI T1S1.2/94-311	Preset Conference Calling Service, DSS1, Stage 3 Description
ANSI T1S1/94-553	ISDN Explicit Call Transfer
ANSI T1S1/LB95-05	ISDN Call Park Supplementary Service
ANSI X3.4 1986	Code for Information Interchange
ANSI X3.16 1976	Character Structure and Character Parity Sense for Serial-by-Bit Data Communication in ASCII
ANSI X3.25 1968	Character Structure and Character Parity Sense for Parallel-by-Bit Data Communication in ASCII

ANSI X3.229	Fiber Distributed Data Interface (FDDI) Station Management

(Copies of American National Standards Institute (ANSI) standards may be obtained from the American National Standards Institute, 1430 Broadway, New York, NY 10018. Draft copies of ANSI standards can be obtained from the Joint Interoperability and Engineering Organization (JIEO), ATTN: JEBBB, Fort Monmouth, NJ 07703-5613.)

2.3.3 International Standards Organization (ISO)/International Electrotechnical Commission(IEC) documents

TR 10000	Information Technology -- Framework and Taxonomy of International Standardized Profiles -- Part 1: Framework, Part 2: Taxonomy of Profiles, and Part 3: Principles and Taxonomy for Open Systems Environment Profiles (Draft)
ISO 3166	Codes for the Representation of Names of Countries
ISO 3309	Information Processing Systems -- Data Communication -- High-Level Data Link Control Procedures -- Frame Structure
ISO 4335	Information Processing Systems -- Data Communication -- High-Level Data Link Control Elements of Procedures
ISO 6523	Data Interchange -- Structure for the Identification of Organizations
ISO 7498	Information Processing Systems -- Open Systems Interconnection -- Basic Reference Model -- X-ref: ITU-T X.200
ISO 7776	Information Processing Systems - Data Communication - High-level Data Link Control Procedures - Description of the X.25 LAPB-Compatible DTE Data Link Procedures
ISO 7809	Information Processing Systems -- Data Communication -- High-Level Data Link Control Procedures -- Consolidation of Classes of Procedures

ISO 8073	Information Processing Systems -- Open Systems Interconnection -- Connection Oriented Transport Protocol Specification -- X-ref: ITU-T X.224
ISO 8208	Information Processing Systems -- Data Communications -- X.25 Packet Level Protocol for Data Terminal Equipment -- X-ref: ITU-T X.25
ISO 8326	Information Processing Systems -- Open Systems Interconnection -- Basic Connection-Oriented Session Service Definition -- See: ITU-T X.215
ISO 8327	Information Processing Systems -- Open Systems Interconnection -- Basic Connection-Oriented Session Protocol Specification -- See: ITU-T X.225
ISO 8348	Information Processing Systems -- Data Communications -- Network Service Definition -- X-ref: ITU-T X.213
ISO 8471	Data Communication -- High-Level Data Link Control Balanced Classes of Procedures -- Data-Link Layer Address Resolution/Negotiation in Switched Environments
ISO 8473	Information Processing Systems -- Data Communications -- Protocol for Providing the Connectionless-Mode Network Service
ISO 8571-1	Information Processing Systems -- Open Systems Interconnection -- File Transfer, Access and Management -- Part 1: General Introduction
ISO 8571-3	Information Processing Systems -- Open Systems Interconnection -- File Transfer, Access and Management -- Part 3: File Service Definition

ISO 8571-4	Information Processing Systems -- Open Systems Interconnection -- File Transfer, Access and Management -- Part 4: File Protocol Specification
ISO 8613-1	Information Processing -- Text and Office Systems -- Office Document Architecture (ODA) and Interchange Format -- Part 1: Introduction and General Principles -- X-ref: ITU-T T.411
ISO 8613-2	Information Processing -- Text and Office Systems -- Office Document Architecture (ODA) and Interchange Format -- Part 2: Document Structures -- X-ref: ITU-T T.412
ISO 8613-4	Information Processing -- Text and Office Systems -- Office Document Architecture (ODA) and Interchange Format -- Part 4: Document Profile -- X-ref: ITU-T T.414
ISO 8613-5	Information Processing -- Text and Office Systems -- Office Document Architecture (ODA) and Interchange Format -- Part 5: Office Document Interchange Format (ODIF) -- X-ref: ITU-T T.415
ISO 8613-6	Information Processing -- Text and Office Systems -- Office Document Architecture (ODA) and Interchange Format -- Part 6: Character Content Architectures -- X-ref: ITU-T T.416
ISO 8613-7	Information Processing -- Text and Office Systems -- Office Document Architecture (ODA) and Interchange Format -- Part 7: Raster Graphics Content Architectures -- X-ref: ITU-T T.417
ISO 8613-8	Information Processing -- Text and Office Systems -- Office Document Architecture (ODA) and Interchange Format -- Part 8: Geometric Graphics Content Architectures -- X-ref: ITU-T T.418

ISO 8632	Computer Graphics Metafile
ISO 8649	Information Processing Systems -- Open Systems Interconnection -- Service Definition for the Association Control Service Element -- See: ITU-T X.217
ISO 8650	Information Processing Systems -- Open Systems Interconnection -- Protocol Specification for the Association Control Service Element -- See: ITU-T X.227
ISO 8802-2	Information Processing Systems -- Local Area Networks -- Part 2: Logical Link Control
ISO 8802-3	CSMA/CD Media Access Control
ISO 8802-4	Token Bus Media Access Control
ISO 8802-5	Token Ring Media Access Control
ISO 8822	Information Processing Systems -- Open Systems Interconnection -- Connection-Oriented Presentation Service Definition -- See: ITU-T X.216
ISO 8823	Information Processing Systems -- Open Systems Interconnection -- Connection-Oriented Presentation Protocol Specification -- See: ITU-T X.226
ISO 8824	Information Processing Systems -- Open Systems Interconnection -- Specification of Abstract Syntax Notation One (ASN.1) -- See: ITU-T X.208
ISO 8825	Information Processing Systems -- Open Systems Interconnection -- Specification of Basic Encoding Rules for Abstract Syntax Notation One (ASN.1) -- See: ITU-T X.209
ISO 8878	Information Processing Systems -- Data Communications -- Use of X.25 to Provide the OSI Connection-Mode Network Service -- See: ITU-T X.223

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ISO 8879	Information Processing Systems -- Standard Generalized Markup Language (SGML)
ISO 8880	Information Processing System -- Open Systems Interconnection -- Protocol Combinations to Provide and Support the OSI Network Service
ISO 8885	Information Processing Systems -- Data Communication High-Level Data Link Control (HDLC) Procedures -- General-Purpose XID Frame Information Field Content and Format
ISO 9040	Information Processing Systems -- Open Systems Interconnection -- Virtual Terminal Service
ISO 9041	Information Processing Systems -- Open Systems Interconnection -- Virtual Terminal (VT) Protocol -- Basic Class
ISO 9069	Information Processing Systems -- SGML Support Facilities -- SGML Document Interchange Format
ISO 9070	Information Processing Systems -- SGML Support Facilities -- Registration Procedures for Public Text
ISO 9314-1	Information Processing Systems -- Fibre Distributed Data Interface (FDDI) -- Part 1: Physical Layer Protocol (PHY) -- See: ANSI X3.148
ISO 9314-2	Information Processing Systems -- Fibre Distributed Data Interface (FDDI) -- Part 2: Token Ring Media Access Control (MAC) -- See: ANSI X3.139
ISO 9314-3	Information Processing Systems -- Fibre Distributed Data Interface (FDDI) -- Part 3: Physical Layer Medium Dependent (PMD) Requirements -- See: ANSI X3.166

ISO 9542	Information Processing Systems -- Telecommunications and Information Exchange Between Systems -- End System to Intermediate System Routing Exchange Protocol for Use in Conjunction with the Protocol for Providing the Connectionless-Mode Network Service
ISO 9595	Information Technology -- Open Systems Interconnection -- Common Management Information Service Definition
ISO 9596-1	Information Technology -- Open Systems Interconnection -- Common Management Information Protocol -- Part 1: Specification
ISO 9646	Open Systems Interconnection -- Conformance Testing Methodology and Framework
ISO DIS 10589	Information Processing Systems -- Intermediate System to Intermediate System Routing Protocols
ISO XXXX	Remote Operations Service Element (ROSE) (Draft)
ISP 10607 (6 Parts)	Information Technology -- International Standardized Profile AFTnn -- File Transfer, Access, and Management (Draft)
ISP 10609 (9 Parts)	Information Technology -- International Standardized Profile TB, TC, TD and TE -- Connection-Mode Transport Service Over Connection-Mode Network Service (Draft)

(Copies of ISO standards may be obtained from the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

2.3.4 Institute of Electrical and Electronics Engineers (IEEE) standards

IEEE 802.1D	MAC Bridges
IEEE P802.1G/1D	Remote MAC Bridge

IEEE 802.6	Distributed Queue Dual Bus (DQDB) Subnetwork of a Metropolitan Area Network (MAN)

(Copies of IEEE standards may be obtained from the Secretary, IEEE Standards Board, Institute of Electrical and Electronics Engineers, Inc., P.O. Box 1331, 445 Hoes Lane, Piscataway, NJ 08855-1331.)

(NOTE: IEEE 802.3, 802.4, and 802.5 are referenced as ISO 8802-3, 8802-4, and 8802-5.)

2.3.5 Requests for comment (RFC) and Internet Activities Board (IAB) standards

RFC 1157	Simple Network Management

(RFCs are available, free of charge, via e-mail using the following address: mailserv@ds.internic.net. Type "send rfcxxxx.txt" in the body.)

2.3.6 Electronic Industries Association (EIA) documents

EIA 232	Interface Between Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange
EIA 422	Electrical Characteristics of Balanced Voltage Digital Interface Circuits
EIA 423	Electrical Characteristics of Unbalanced Voltage Digital Interface Circuits

(Copies of EIA standards may be obtained from ANSI or EIA, Electronic Industries Association, Engineering Department, 2001 Eye Street, Northwest, Washington, D.C. 20006.)

2.3.7 Corporation for Open Systems (COS) documents

COS VTC001	Industry Profile for Video Teleconferencing

(Copies of this document may be obtained from JIEO, Center for Standards, Fort Monmouth, NJ 07703-5613, Attention: JEBBC, Mr. Klaus Rittenbach.)

2.3.8 Bellcore documents

TR-TSV-000772	Generic System Requirements in Support of Switched Multi-megabit Data Service

(Copies of Bellcore documents may be requested from Bellcore Customer Service, 8 Corporate Place, Piscataway, NJ 08854-4156, telephone: 1-800-521-2673.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. In addition, whenever a DoD Standardized Profile (DSP) and the associated International Standardized Profile (ISP) are listed, the DSP takes precedence.

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3. DEFINITIONS

3.1 Acronyms and abbreviations. The acronyms and abbreviations used in this standard are defined as follows:

AAL	ATM adaptation layer
ABR	available bit rate
ACK	acknowledgment
ACSE	association control service elements
A-D	analog-to-digital
ADP	automatic data processing
ADPCM	adaptive differential pulse-code modulation
AFI	authority and format identifier
AITS	Adopted Information Technology Standard
AJ	anti-jam
ALE	automatic link establishment
ANSI	American National Standards Institute
ARIDPCM	Adaptive Recursive Interpolated Differential PCM
ASCII	American Standard Code for Information Interchange
ASN.1	abstract syntax notation 1
ATDL-1	Army Tactical Data Link 1
ATM	asynchronous transfer mode
B-Channel	bearer channel
BCI	bit count integrity
BER	bit error ratio
B-ISDN	broadband-ISDN
BNZS	bipolar with N-zero substitution
bps	bit(s) per second
BPSK	binary phase-shift keying
BRI	basic rate interface
BT	broadband terminal
CBR	constant bit rate
CC	country code
CCIR	International Radio Consultation Committee
CCITT	International Telegraph and Telephone Consultative Committee (now referred to as ITU-T)
CDMA	code-division multiple access
CELP	code-excited linear prediction
CIM	Corporate Information Management
CLIP	Calling Line Identification Presentation
CLIR	Calling Line Identification Restriction
CLNS	connectionless network service
CLP	cell loss priority
CMIP	Common Management Information Protocol
CMIS	Common Management Information Services
CNR	combat net radio
COMPUSEC	computer security
COMSEC	communications security

CONS	connection-oriented network service
CONUS	continental United States
COTS	connection-oriented transport service (also, commercial off-the-shelf)
CPCS	common part convergence sublayer
CRBS	cell-relay bearer service
CRC	cyclic redundancy check
CS	convergence sublayer
CSMA/CD	carrier sense multiple access/collision detection
CSN	circuit-switched network
CVSD	continuously variable slope delta
C3I	command, control, communications, and intelligence
C4I	command, control, communications, computers, and intelligence
D	data
DAMA	demand-assignment multiple access
DBMS	database management system
D-channel	16- or 64-kbps channel for signaling and data
dc	direct current
DCA	Defense Communications Agency (now DISA)
DCAC	DCA circular
DCC	data country code
DCC	data communications channel
DCE	data circuit-terminating equipment
DCP	data communications protocol
DCPS	DoD Communications Protocol Standards
DCS	Defense Communications System
DDN	Defense Data Network
DEQPSK	differentially encoded quadrature phase-shift keying
DII	Defense Information Infrastructure
DIR	directory
DIS	Defense Information System; Draft International Standard
DISA	Defense Information Systems Agency (formerly DCA)
DISN	Defense Information System Network
DL	data link
DNS	Domain Name Service
DoD	Department of Defense
DoDISS	Department of Defense Index of Specifications and Standards
DPI	data processing installation
DQDB	distributed queue dual-bus
DS	directory service
DSN	Defense Switched Network
DS1	Digital Interface Rate 1 (1.544 Mbps)
DS2	Digital Interface Rate 2 (6.312 Mbps)
DS3	Digital Interface Rate 3 (44.736 Mbps)
DSP	domain-specific part

DSP	DoD standardized profile
DSS1	Digital Subscriber Signaling System Number 1
DTE	data terminal equipment
DTH	down-the-hill
DTMF	dual-tone multifrequency
ECCM	electronic counter-countermeasures
EDI	Electronic Data Interchange
EFS	error-free second
EHF	extremely high frequency
EIA	Electronic Industries Association
EMC	electromagnetic compatibility
EMSEC	emission security
ES	end system
ESF	extended superframe
ETSI	European Telecommunications Standards Institute
EW	electronic warfare
FAX	facsimile
FDDI	Fiber Distributed Data Interface
FEC	forward error correction
FED-STD	federal standard
FIPS	Federal Information Processing Standard
FRM	frame relay mode
FSK	frequency-shift keying
FTAM	file transfer, access, and management
FTP	File Transfer Protocol
Gbps	gigabit(s) per second
GFC	generic flow control
GHz	gigahertz
GNMP	Government Network Management Profile
GOSIP	Government Open Systems Interconnection Profile
GSA	General Services Administration
GSM	Special Mobile Group
H-Channel	high-rate channel
H_0-Channel	384 kbps
H_{12}-Channel	1,472 kbps
HDB3	high-density bipolar with a maximum of 3 consecutive zeros
HDLC	high-level data link control
HDTV	high-definition television
HEC	header error check
HF	high frequency
HRC	hypothetical reference circuit
HRX	hypothetical reference connection
Hz	hertz
I	imagery
IAB	Internet Activities Board

ICD	international code designator
IDI	initial domain identifier
IDP	initial domain part
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IP	internet protocol
IPMS	interpersonal message service
IPR	IP router
IRAC	Interdepartment Radio Advisory Committee
IS	intermediate system
ISDN	Integrated Services Digital Network
ISO	International Organization for Standardization
ISP	International Standardized Profile
ITU	International Telecommunications Union
ITU-T	ITU-Telecommunication Standardization Sector (formerly CCITT)
IWG	interim working group
JCS	Joint Chiefs of Staff
JIEO	Joint Interoperability and Engineering Organization
JITC	Joint Interoperability Test Center
JPEG	Joint Photographic Experts Group
JTIDS	Joint Tactical Information Distribution System
kbps	kilobit(s) per second
kHz	kilohertz
km	kilometer(s)
LAN	local area network
LAP	link access procedure
LAPB	LAP balanced
LAPD	LAP on the D-channel
LDR	low data rate
LF	low frequency
LLC	logical link control
LME	layer management entity
LOS	line-of-sight
LPC	linear predictive coding
MAC	media access control
MAN	metropolitan area network
MAU	medium attachment unit
Mbps	megabit(s) per second
MCEB	Military Communications-Electronics Board
MDR	medium data rate
MF	medium frequency
MHS	message-handling service
MHz	megahertz
MIB	management information base

MIDS	Multi-Functional Information Distribution System
MILDEP	military department
MIL-HDBK	military handbook
MIL-STD	military standard
MLPP	Multi-level Precedence and Preemption
MM	military messaging
MMS	Military Messaging Service
ms	millisecond(s)
MSP	message security protocol
MSR	message storage and retrieval
MTBF	mean time between failures
MTBPM	mean time between preventive maintenance
MTP	message transfer part
MTTR	mean time to repair
mW	milliwatt(s)
MWI	message waiting indicator
n	integer
NACK	nonacknowledgment
NACSEM	National COMSEC Engineering Memorandum
NACSIM	National COMSEC Information Memorandum
NATO	North Atlantic Treaty Organization
NCC	network control center
NDI	nondevelopmental item
NE	network element
NI	nationality identifier
N-ISDN	narrowband ISDN
NIST	National Institute of Standards and Technology
NITF	National Imagery Transmission Format
NITFS	NITF standard
NLSP	network-layer security protocol
NM	network management
NNI	network-node interface
NRI	net radio interface
NSA	National Security Agency
NTIS	National Technical Information Service
OC	optical carrier level
ODA	Office Document Architecture
OS	operating system
OSI	Open Systems Interconnection
OQPSK	offset quadrature phase-shift keying
PBX	private branch exchange
PCM	pulse-code modulation
PCS	personal communications services
PDN	public data network
PDU	protocol data unit
pFS	proposed Federal Standard
PG/6	Project Group 6
PHY	Physical Layer Protocol

PLP	packet level protocol
PLRS	Position Location Reporting System
PMD	Physical Layer Medium Dependent
POH	path overhead
POSIT	Profiles for Open Systems Internetworking Technologies
ppm	part(s) per million
PRI	primary rate interface
PSK	phase-shift keying
PSN	packet-switched network
PSTN	public switched telephone network
PTI	payload type identifier
PTS	personal telecommunications service
QOS	quality-of-service
R	radio
RES	reserved
rf	radio frequency
RFC	request for comment
RM	reference model (OSI)
ROSE	remote operations service element
SAAL	AAL for signaling
SAR	segmentation and reassembly
SATCOM	satellite communications
SATURN	Second-Generation Anti-Jam Tactical UHF Radio for NATO
SCCP	signaling connection control part
SDH	synchronous digital hierarchy
SDLS	satellite data link standard
SGML	Standard Generalized Markup Language
SHF	super high frequency
SINCGARS	Single-Channel Ground and Airborne Radio System
SMFA	system management functional area
SMT	station management
SMTP	Simple Mail Transfer Protocol
SNE	subscriber network element
SNMP	Simple Network Management Protocol
SOH	synchronous optical hierarchy
SONET	synchronous optical network
SOQPSK	shaped offset quadrature phase-shift keying
SSAP	session service access point
SSCF	service-specific coordination function
SSCOP	service-specific connection-oriented protocol
SSCS	service-specific convergence sublayer
SS7	Signaling System Number 7
STANAG	standardization agreement
STM	synchronous transfer mode
STM-N	synchronous transport module - level N
STS	synchronous transport signal

TAC02	Tactical Communications Protocol 2
TADIL	tactical digital information link
TAFIM	Technical Architecture Framework for Information Management
TBD	to be determined
TC	transmission convergence
TCAP	transaction capabilities application part
TCP	transmission control protocol
TDM	time-division multiplexing
TDMA	time-division multiple access
TE	terminal equipment
TEMPEST	compromising emanations
TIA	Telecommunications Industry Association
TLSP	transport-layer security protocol
TP4	Transport Protocol class 4
TPDU	transport protocol data-unit
TR	technical report
TRANSEC	transmission security
TRI-TAC	Tri-Service Tactical Communications
TTY	teletypewriter
UDP	user datagram protocol
UHF	ultra high frequency
ULF	ultra low frequency
UMTS	Universal Mobile Telecommunications System
UNI	user-to-network interface
USAT	ultra-small aperture terminal
UTC	coordinated universal time
V	voice
VBR	variable bit rate
VCI	virtual channel indicator
VHF	very high frequency
VOX	voice-operated transmit
VPC	virtual path connection
VPI	virtual path indicator
VT	virtual terminal
VTC	video teleconferencing
WAN	wide area network
WARC	World Administrative Radio Conference
WG	working group
WNE	wide-network element
2B1Q	two binary, one quaternary

3.2 Definitions of terms. Definitions of terms used in this MIL-STD shall be as specified in Federal Standard (FED-STD)-1037. Those definitions unique to information systems, and not defined in FED-STD-1037, are provided in this section.

Broadband terminal (BT): User equipment at the location where the user-to-network interface (UNI) terminates. The BT may be a single

user equipment or it may act as an aggregator of other tributary devices.

Functional profiles: A profile is defined in TR 10000 as a "set of one or more base standards, and, where applicable, the identification of chosen classes, subsets, options, and parameters of those base standards, necessary for accomplishing a particular function." The term *profile*, as used in MIL-HDBK-829, has the same meaning, as does the term *functional profile*. Profile and functional profile are used interchangeably.

Generic flow control (GFC): A bit sequence in the asynchronous transfer mode (ATM) cell header that is intended to control traffic flow into the network, to support different quality-of-service (QOS). To date, the GFC has not been defined and remains an unused field. In the interim, the field shall be all zeros.

Local-network elements: Elements that make up a base information-transfer utility for strategic users or a tactical information-transfer utility for tactical users. They include such elements as circuit and packet switches, and transmission equipment. They may also include metropolitan-area networks (MAN).

Path overhead: Overhead assigned to and transported with the SONET payload. It provides for communications between the point of creation of a SONET payload and its point of disassembly.

Reference point A: The interface between subscriber-network elements and local-network elements.

Reference point B: The interface between local-network elements and wide-network elements.

Reference point B (NATO): The interface between U.S. network elements and NATO network elements.

Strategic user: A person, organization, or other entity (including a computer or computer system) not assigned as a tactical user.

Subscriber-network elements: Elements such as terminal equipment, end systems, intermediate systems, local-area networks, metropolitan-area networks, and radio networks normally considered to be provided by the subscriber.

Tactical user: A person, organization, or other entity (including a computer or computer system) in support of a joint task force who employs the services provided by a tactical telecommunications system, or by a tactical information-processing system, for transfer of information to others.

Telnet: The virtual terminal (VT) protocol in the Internet suite of protocols that allows users of one host to log into a remote host and interact as normal terminal users of that host. The

Telnet protocol is specified in IAB STD-8 (and for OSI networks in ISO 9040/9041).

Virtual Channel Indicator (VCI): Defines the explicit cell channel identification at the user-to-network interface (UNI) and network-to-node interface (NNI).

Virtual Path Indicator (VPI): Defines the explicit cell path identification at the UNI and NNI.

Wide-network elements: Elements, such as circuit switches, packet switches, and transmission equipment, that form the Defense Communications System (DCS) and public switched telephone networks (PSTN).

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4. GENERAL REQUIREMENTS

4.1 System requirements. The general system requirements in this section affect the design of subscriber-network elements (information sources, sinks, and subscriber-owned networks), local-network elements, and wide-network elements, as shown in Figure 1. The term *Defense Information System* (DIS), in Figure 1, is synonymous with the Defense Information Infrastructure (DII), except DIS does not include the operations and support staff and facilities. Figure 2 [exhibit 1-2 in the Defense Information System Network (DISN) Architecture], depicts the DII/DISN relationship, including the transform/transfer functions and information use/creation functions that are part of the terminal/computing elements. All systems fielded to satisfy Department of Defense (DoD) requirements shall comply with applicable information technology standards in the DoD Technical Reference Model, which is Volume #7 of the *Technical Architecture Framework for Information Management (TAFIM)*. Use/creation functions are addressed in detail in the DoD Technical Reference Model, and the information-transfer portion is addressed by referencing this MIL-STD.

4.1.1 End-to-end digital service. All signals entering the local- and wide-network elements shall be digital and shall remain in a digital form until the signals exit the local network at reference point A. Analog-to-digital and digital-to-analog conversion, when required, shall occur in terminal equipment or in terminal adapters. The network elements shall preserve bit count integrity (BCI) through the aggregate of network elements.

4.1.2 Circuit-switched services

4.1.2.1 Signaling. The DISN shall provide for user-to-network and user-to-user signaling, as described in 4.1.2.1.1 to 4.1.2.1.3.

4.1.2.1.1 Network-node signaling. Common-channel signaling shall be employed in local networks and wide networks. For tactical information-transfer systems, interswitch common-channel-signaling messages shall comply with MIL-STD-188-256. For base information-transfer systems and wide networks, interswitch common-channel-signaling messages shall comply with ANSI T1.111, T1.112, T1.113, and T1.114 for Signaling System Number 7 (SS7), as modified, to provide the military enhancements described in mandatory Appendix A, titled *DSN7 Common Channel Signaling*. For broadband networks, signaling messages shall comply with ITU-T Q.2761 to Q.2764, and other applicable standards and implementation agreements as they become available.

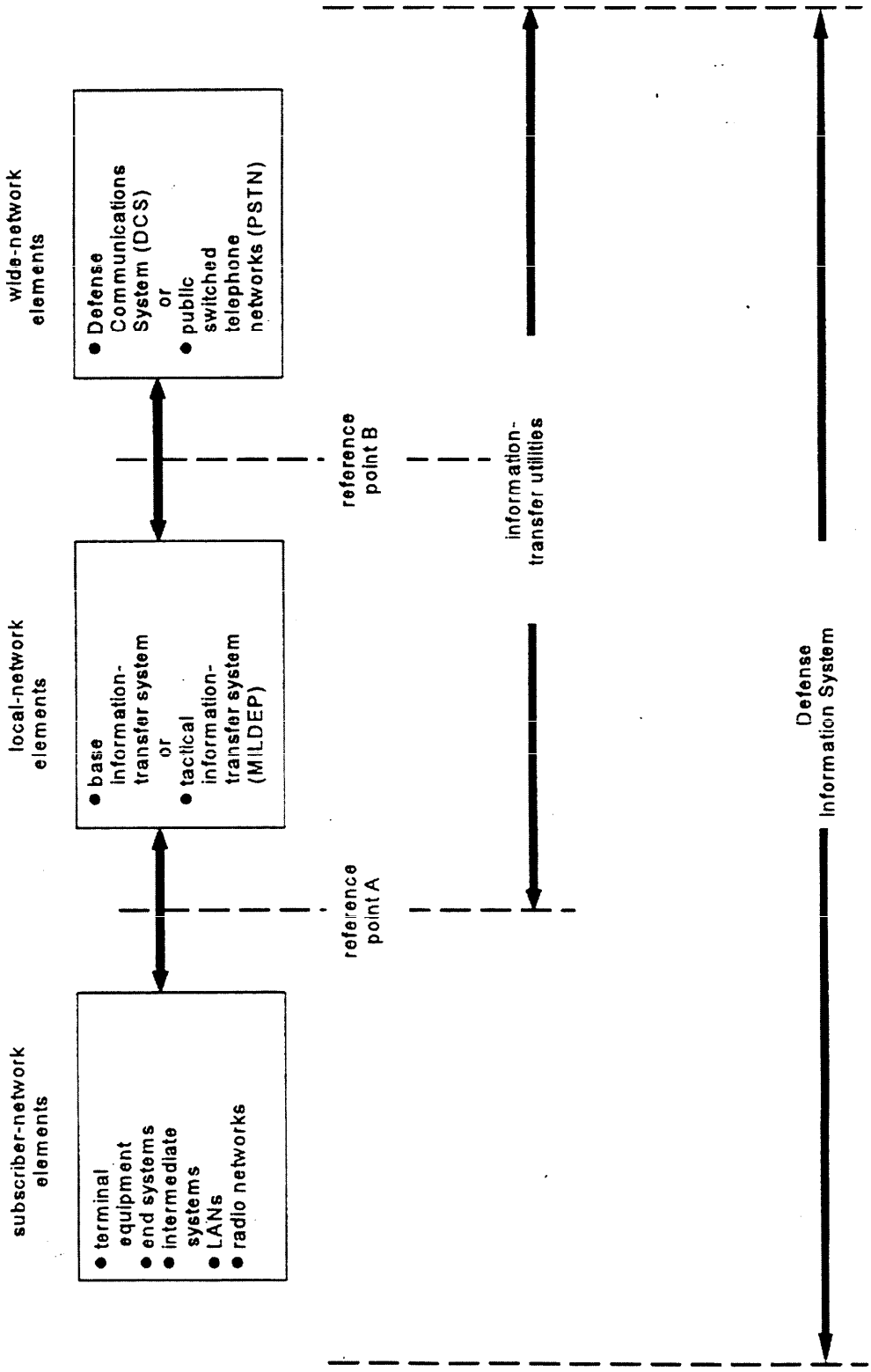


FIGURE 1. DIS framework.

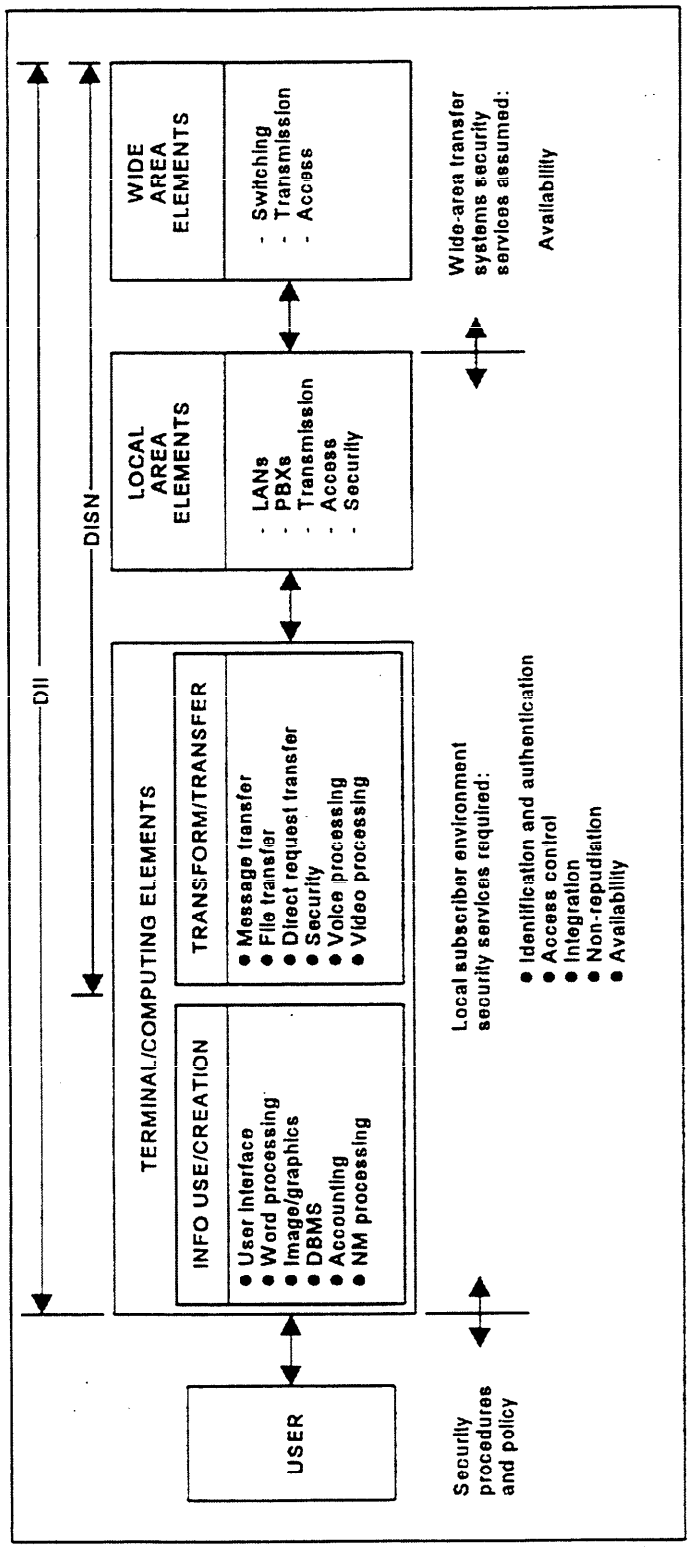


Figure 2. DISN relationship to DII.

4.1.2.1.2 User-to-network signaling

a. Common-channel signaling shall be employed at the user-to-network interface in base information-transfer systems. User-to-network signaling messages shall comply with the following ANSI standards:

(1) ANSI T1.602

(2) ANSI T1.607

(3) ANSI T1.608

(4) ANSI T1.610

b. In-band signaling shall be employed at the user-to-network interface in tactical information-transfer systems. User-to-network signaling messages shall comply with MIL-STD-188-256.

c. For asynchronous transfer mode (ATM) networks, user-to-network signaling messages shall share the same transmission facility used to carry traffic. User-to-network signaling messages shall comply with ITU-T Q.2931.

4.1.2.1.3 User-to-user signaling. User-to-user signaling is the control information exchanged between users' terminal equipment. This information may be transmitted in the signaling and the information bearer channels, but shall be transparent to network elements. The user-to-user information element is used to establish end-to-end encrypted calls between tactical and strategic users, as defined in MIL-STD-188-105.

4.1.3 Packet data service. Packet data service may be provided by different types of packet networks including X.25 packet-switched networks, LANs, and IP Router networks. Figure 3 illustrates how TCP/IP will be used to achieve interoperability between DTEs in different networks.

4.1.3.1 DTE direct access. Packet data services may be delivered by means of data terminal equipment (DTE) directly connected to packet-switched networks. X.25 packet-switched networks shall conform to the Internet Transport Profile for DoD communications, as specified in MIL-STD-2045-14502-3.

4.1.3.2 LAN access. For LAN access, DTEs must match the LAN physical layer as well as the layer 2 sublayers, the Logical Link Control (LLC) and Media Access Control (MAC). DTEs connected to the LAN may exchange packets with other DTEs connected to the

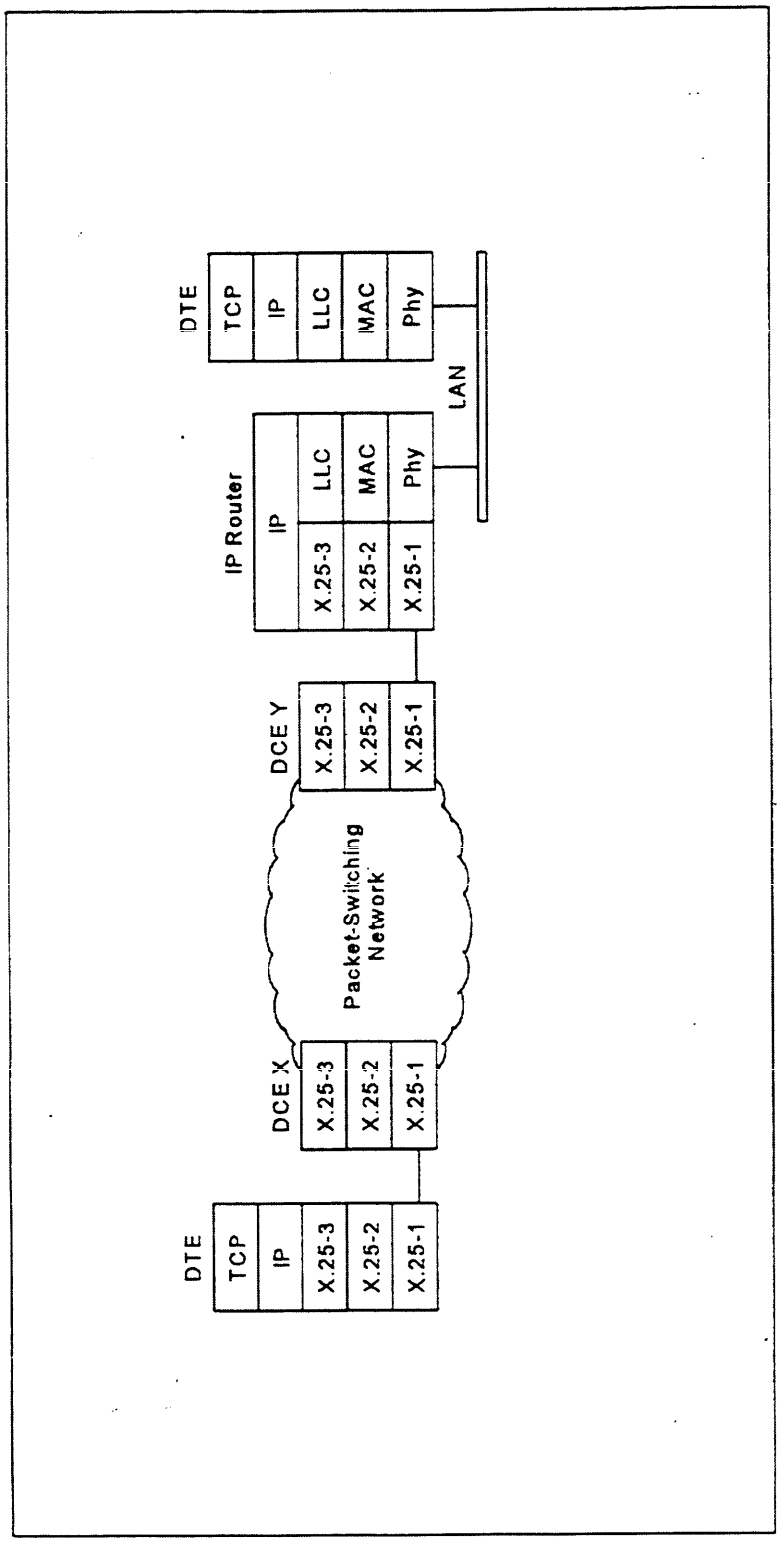


FIGURE 3. Internetworking using TCP/IP.

same LAN. LAN access shall conform to the Internet Transport profiles for DoD communications, as specified in MIL-STD-2045-14502-4 for media-independent requirements and in MIL-STD-2045-14502-5 for media-dependent requirements.

4.1.3.3 DTE interoperability. For DTEs to connect to other DTEs on different subnetworks, the transmission control protocol/internet protocol (TCP/IP) suite, as specified in MIL-STD-2045-14502-1, shall be used. TCP is a connection-oriented transport protocol that ensures reliable end-to-end data service. IP is a connectionless internet protocol that enables interconnection of subnetworks into an internetwork, independent of any particular subnetwork technology, and is used in gateways and routers between subnetworks. The TCP/IP suite includes common applications such as electronic mail, file transfer, and remote login, to aid DTE interoperability.

4.1.3.4 IP router networks. DDN X.25 networks are being phased out and replaced by IP router networks. IP routers perform routing and gateway functions needed to pass packets between different subnetworks and other routers. IP addresses include the address for the subnetwork and DTEs on that subnetwork; therefore, IP routers route packets based on the destination subnetwork address, not the destination DTE address. IP routers can exist at any place within the DII as either interior or exterior gateways. For the purpose of routing, a group of networks and gateways controlled by a single administrative authority is called an autonomous system, using interior gateway protocols. Gateways between autonomous systems use exterior gateways protocols.

4.1.3.5 IP addressing. Currently, IP provides only 32 bits of address space and is facing an inability to provide unique IP addresses to all entities that require them. Work is ongoing on a next-generation protocol (IPng) to resolve this issue.

4.1.3.6 ATM networks. In the future ATM networks will provide the transport service between IP routers. IP datagrams will be segmented into ATM cells at the sending ATM-network interface and reconstructed from ATM cells at the receiving ATM-network interface.

4.1.3.7 Switched multi-megabit data service. Switched multi-megabit data service (SMDS), a Bellcore proprietary development, provides connectionless datagram transfer service. The SMDS variable-length data units are segmented into 53-byte cells for transmission. SMDS addressing uses the ITU-T E.164 numbering plan. Open-commercial standards for the SMDS subscriber network interface have not been developed. Details are provided in Bellcore TR-TSV-000772.

4.1.3.8 Point-to-point access. For point-to-point access, DTEs must match the physical layer as well as the data link layer. Point-to-point access shall conform to the internet transport profile for DoD communications, as specified in MIL-STD-2045-14502-2.

4.1.4 Personal communications services (PCS). *Personal communications services* is defined as a set of capabilities that allows some combination of terminal mobility, personal mobility, and service profile management. Figure 4 shows all system components needed to support PCS. The DISN shall implement personal communications services using the currently emerging commercial standards.

4.1.4.1 Terminal mobility. Terminal mobility is based on wireless access to the network. It allows a mobile terminal, which moves with its user, to gain access to the network from different locations. To support terminal mobility, the network must be able to identify calling wireless terminals and locate called wireless terminals. In Figure 4 terminal, mobility is represented by the vehicular and hand-held terminals shown on the left. Terminal mobility shall be provided initially within each base communications and Joint Tactical Headquarters area. Eventually, global terminal mobility shall be provided across these areas, consistent with the commercial standards for Future Public Land Mobile Telecommunication Systems and Universal Personal Telecommunications.

4.1.4.1.1 Wireless access. Wireless access involves the wireless link (air interface) between the mobile terminal and the base station (radio port), management of the wireless links provided by several base stations by a central base station controller, and the interface between the base station controller and the switched network. In Figure 4, wireless access is represented by the base stations and the base station controller shown in the center.

4.1.4.2 Personal mobility. Personal mobility allows users of telecommunication services to gain access to these services from any convenient terminal with which they choose to associate themselves. To support personal mobility, the network must be able to distinguish between terminal and personal identifiers; to keep track of current user-terminal associations, user locations, services authorized to the user, and service capabilities of the terminals; and to maintain the user's call/service account. Personal mobility may be provided by either wireline or wireless terminals. In the lower right of Figure 4, personal mobility is represented by the two fixed terminals and the user moving between them. Personal mobility shall be provided globally across all DISN terminals.

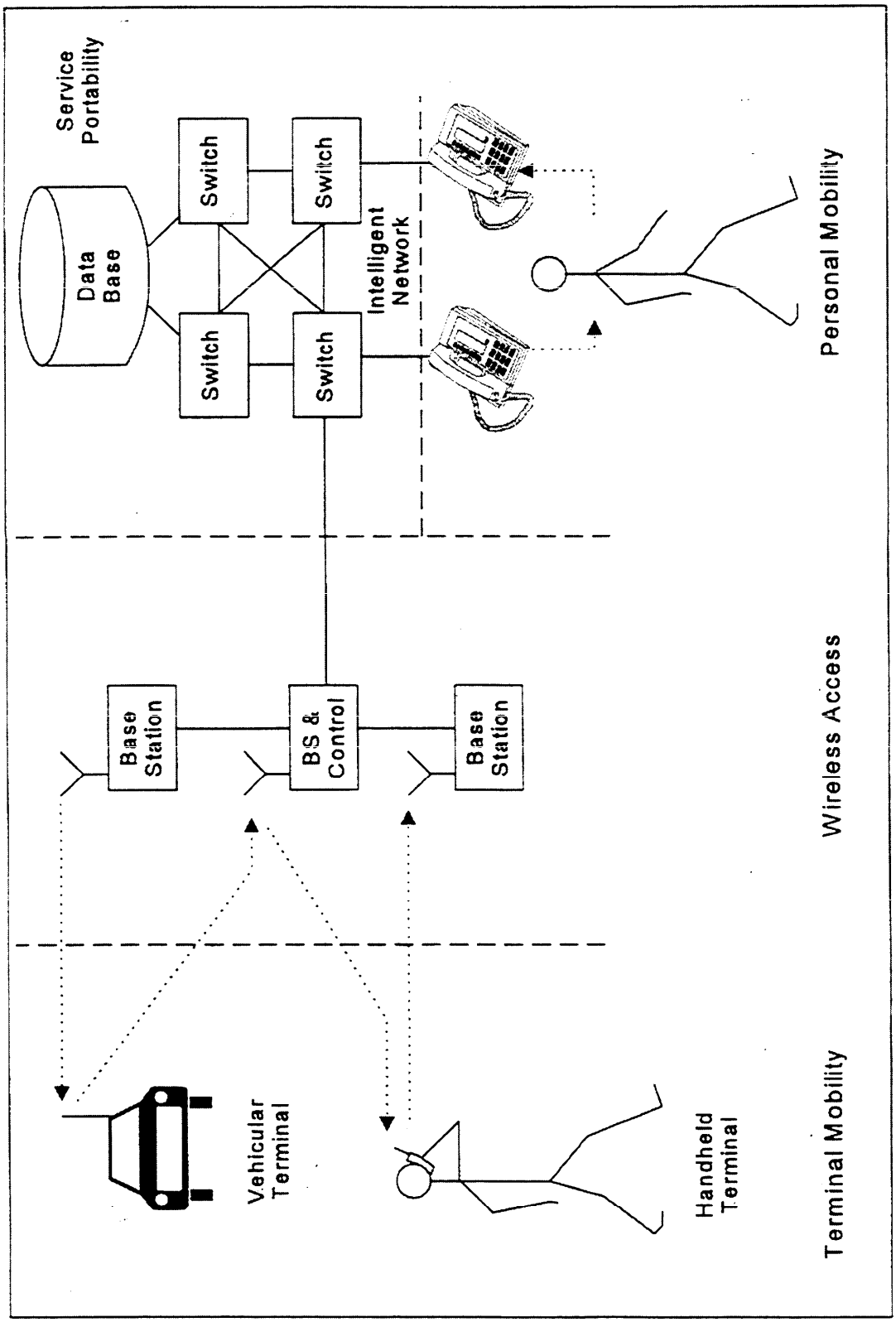


FIGURE 4. Personal communications services.

4.1.4.3 Universal access. Universal access is the combination of both terminal and personal mobility, including the ability of wireless terminal users to move between different wireless access areas. The term *intelligent network* implies the capabilities needed to support universal access. An intelligent network must contain a central or distributed database and perform service profile management. Service profile management maintains the home (normal) location of users, the location they are currently visiting, the identity of the terminal with which they are currently associated, their currently authorized services, their service use records, and the service capabilities of each terminal and wireless access area. Additionally, the signaling system used in an intelligent network must be able to rapidly interrogate this database without causing undue delays or a substantial increase in communications traffic. The key components of an intelligent network are shown in the upper right of Figure 4. The DISN shall be enhanced, commensurate with commercial developments, to support intelligent network requirements and universal access.

4.1.5 Dedicated circuits. The DISN shall be able to provide dedicated circuits at 64-kbps, 384-kbps, and 1.544-Mbps rates. These circuits shall be provided by commercially leased lines or by multiplexing existing channels into dedicated through groups. The layer 1 interface shall conform to applicable portions of 5.1.1.1.1 for the 64-kbps rate, and 5.1.2.1.1 for both the 384-kbps and 1.544-Mbps rates.

4.1.6 Subscriber services. Local-network elements shall provide telecommunications subscriber services for circuit-switched voice, circuit-switched data, and packet-switched data. Services supported include voice telephony, data transmission, facsimile, record traffic, video, and multimedia communications.

4.1.6.1 Voice services. The aggregate of tactical-network elements, wide-network elements, and base-level network elements shall provide the capability for both nonsecure and end-to-end encrypted voice calls between subscribers at a military base and subscribers in a tactical system. A discussion of nonsecure voice calls is described in 4.1.6.1.2; end-to-end encrypted voice calls are discussed in 4.1.6.1.3.

4.1.6.1.1 Voice digitization. The following voice digitization algorithms shall be used:

a. Base information-transfer systems shall use 64-kbps PCM (mu-law companding), as defined in ITU G.711, Tables 2a and 2b.

b. Tactical information-transfer systems shall be able to interface either directly or via a switch, using 16-kbps continuously variable slope delta (CVSD) modulation, as defined in MIL-STD-188-113, the section titled *CVSD modulation*. Tactical

systems employing 32-kbps digital loops shall double-sample 16-kbps CVSD signals to achieve joint interoperability.

c. Narrowband voice subscribers connected to the local network through high frequency (HF) radio, special electronic counter-countermeasures (ECCM) radios, or other narrow-bandwidth facilities shall use 2.4-kbps linear predictive coding (LPC), as defined in MIL-STD-188-113, the section titled *LPC*; or 4.8-kbps code-excited linear prediction (CELP), as defined in FED-STD-1016.

d. Digital voice encoders employing 32-kbps adaptive differential pulse-code modulation (ADPCM) shall conform to ITU-T G.721.

4.1.6.1.2 Nonsecure voice service. The Integrated Services Digital Network (ISDN) and tactical telephone terminals use different voice-encoding algorithms and different bit rates dictated by the bandwidth of the transmission equipment used in DoD networks. Base telephone terminals use 64-kbps PCM voice encoding. Tactical telephone terminals use 16-kbps CVSD (optionally 32-kbps CVSD) voice encoding. For nonsecure voice calls between subscribers at a military base and subscribers in a tactical network, transcoding will be done at the internetwork gateway to convert from 64-kbps PCM to 16-kbps CVSD (optionally 32-kbps CVSD), and from 16-kbps CVSD (optionally 32-kbps) to 64-kbps PCM, as described in MIL-STD-188-105, the two sections titled *Transcoding*.

4.1.6.1.3 End-to-end encrypted voice service. End-to-end encrypted voice service between tactical and strategic networks shall be available if the subscriber terminals contain a common voice-encoding algorithm and a common crypto algorithm, and are capable of mode negotiation. End-to-end encrypted voice calls shall be treated as a data service. BCI shall be preserved to maintain cryptographic synchronization between calling and called secure-voice terminals. End-to-end encrypted voice calls between base-level and tactical users may occur at any standard bit rate up to and including 16 kbps (optionally 32 kbps), as described in MIL-STD-188-105, the section titled *End-to-end encrypted telephone service*. The network elements shall allow switching from data-to-voice and voice-to-data to occur at the terminals.

NOTE: The 32-kbps CVSD option will work only for special cases in which calls do not transit subnetworks that use 16-kbps channels.

4.1.6.1.3.1 Mode negotiation. New telephone terminals used in ISDN networks will be able to negotiate, during the call-establishment phase, to determine if a common voice-encoding mode exists. The new telephone terminal shall comply with MIL-STD-188-105, the section titled *New terminal*, and addressed by the National Security Agency (NSA) in their Secure Terminal Equipment

Program. When used in ISDN networks, mode negotiation shall be accomplished using the user-to-user information element, as described in MIL-STD-188-105, the section titled *Call-establishment phase*.

4.1.6.1.3.2 Interoperable modes. New telephone terminals, as defined in MIL-STD-188-105, shall have at least two voice-encoding modes. One will be PCM, which will be used in ISDN networks, as defined in FIPS-PUB 182. Another will be CVSD-16, which will be used for end-to-end encrypted calls between tactical and strategic users. Other voice-encoding schemes may be 32-kbps ADPCM, 2.4-kbps LPC-10, and 4.8-kbps CELP.

4.1.6.1.3.3 End-to-end encrypted voice service in ATM networks. End-to-end encrypted voice service shall use AAL 1 CBR service (see 5.4.3.3.3) for access to and exit from ATM networks, to maintain BCI. The AAL 1 protocol may be implemented in either the telephone terminal or a terminal adapter. The terminal adapter may be located in the ATM switch.

4.1.6.1.4 Rate adaptation

a. Information sources, linked to a strategic-local network, that operate at rates of 600, 1200, 2400, 4800, 9600, 16000, 19200, or 32000 bps, shall be rate-adapted to a 64-kbps channel. The rate adaptation of bit rates up to 32 kbps shall use the multi-stage approach defined in ITU-T V.110, the section titled *Adaptation of V-series data signaling rates to the intermediate rates*. With this approach, rates of 4.8 kbps and below are mapped to 8 kbps, 9.6 kbps is mapped to 16 kbps, and 19.2 kbps is mapped to 32 kbps. Rate adaptation of 8-, 16-, and 32-kbps signals shall be rate-adapted in accordance with the following procedure, as documented in ITU-T I.460, the section titled *Rate adaptation of 8-, 16-, and 32-kbps streams*:

(1) The 8-kbps stream occupies bit position 1.

(2) The 16-kbps stream occupies bit positions 1 and 2.

(3) The 32-kbps stream occupies bit positions 1, 2, 3, and 4.

(4) All unused bit positions shall be set to "1."

(5) The order of bit transmission of the subrate stream shall be identical before and after rate adaptation.

b. Information sources, linked to a tactical network, that operate at rates of 75, 600, 1200, 2400, 4800, or 9600 bps, shall be rate-adapted to a 16-kbps channel, as described in MIL-STD-188-216, the section titled *Multisampling*.

4.1.6.2 Data services. The aggregate of tactical-network elements, wide-network elements, and base-level network elements shall provide the capability for end-to-end data service such as interactive, video, and imagery data. The data service shall be supported by internet protocols; open system interconnection protocols; or any combination of both protocols over packet, frame, and cell switching, or router networks. The separation of traffic by classification shall be provided by (1) physical separation through network segmentation; (2) end-to-end encryption, wherein the keys associated with the encryption are protected as a minimum to the same level as the classification of the traffic that they handle; or (3) a combination of both. Authentication of nonsecure dial-up subscribers shall be provided by NSA- approved non-forgeable dynamic-token-based authentication technology. Authentication of secure dial-up subscribers and encryption of their traffic shall be provided through the use of NSA-approved secure terminal equipment (STE).

4.1.6.3 Video services. Circuit-switched networks shall support video services by allocating multiple channels (or timeslots) as required to provide sufficient bandwidth for video signals. This service will support nx64-kbps operation (where 1<n<23) for base networks and mx16-kbps (where 1<m<16) for tactical networks. The bandwidth provided for video services between strategic and tactical networks may be limited by the tactical networks. The procedure for converting 64-kbps to 16-kbps channels is provided in MIL-STD-188-105 w/notice of change.

4.1.6.4 Multimedia communications. Network access and signaling functions shall support multimedia services. Multimedia services provide the capability to manipulate and manage information consisting of text, graphics, images, video, and audio. These services can be used directly by mission area applications and by other support applications, as described in the TAFIM. Addition and deletion of types of information during an active multimedia session shall be supported.

4.1.6.5 Supplementary services. A network supplies supplementary services in addition to its basic services. The generic procedures applicable to the control of supplementary services at the user-to-network interface are defined in ANSI T1.610. The mandatory supplementary services, which are applicable to both fixed and deployed networks, are discussed in 4.1.6.5.1 to 4.1.6.5.3. The optional supplementary services are discussed in 4.1.6.5.4 to 4.1.6.5.17.

4.1.6.5.1 Multi-level Precedence and Preemption. The Multi-level Precedence and Preemption (MLPP) service provides a prioritized call-handling service. This service has two parts: precedence and preemption. Precedence involves assigning a priority level to a call. Preemption involves the seizing of resources, which are in use by a call of lower precedence, by a higher-level precedence call in the absence of idle resources.

The MLPP service is a network provider's option applicable to a domain of the network, that is, all subscribers and the network and access resources that belong to the domain. Connections and resources belonging to calls from MLPP subscribers shall be marked with a precedence level and domain identifier and shall be preempted only by calls of a higher precedence from MLPP users in the same domain. Connections and resources belonging to calls from non-MLPP users and users from other MLPP domains shall not be preempted. The maximum precedence level of a subscriber will be set by the service provider, based on the subscriber's need. The subscriber may select a precedence level up to and including the maximum subscribed-to precedence level on a per-call basis. The MLPP service shall be mandatory in DoD networks (both fixed and deployed) and shall comply with ANSI T1.619. For calls to subscribers in existing deployed (tactical) networks that comply with Tri-Service Tactical Communications (TRI-TAC) specifications, the MLPP service shall comply with MIL-STD-188-105.

4.1.6.5.2 Preset Conference. The Preset Conference service allows the served user to quickly establish a conference call with a predetermined list of conferees stored in the network. All of the call attempts are done in parallel. This service is defined in ANSI Draft T1S1.2/94-311.

4.1.6.5.3 Hotline. The Hotline service allows a user to automatically initiate a call, act as a receiver for Hotline service calls, or do both. Three subscription options exist for the Hotline service: Protected Hotline Calling service, Unprotected Hotline Calling service, and Protected Hotline Receiving service. When subscribing to the Protected Hotline Calling service, the user shall initiate calls only to the predetermined called party. In addition, the Protected Hotline Calling service user will not receive calls from any third party and may receive calls only from the predetermined called party, if that predetermined called party is also a Hotline service user. When subscribing to the Unprotected Hotline Calling service, the user shall initiate calls only to the predetermined called party of the Hotline service call, who may or may not be another Hotline service user; however, the Hotline service user may receive calls from third parties in addition to calls from the called party of the Hotline service call. A protected Hotline Receiving service user may receive calls only from predetermined Hotline Calling service user(s); however, the user may initiate calls to other users. This service is defined in ANSI Draft T1S1.2/91-408.

4.1.6.5.4 Call Waiting. The Call Waiting service permits a subscriber to be notified of an incoming call with an indication that no interface information channel is available. The subscriber then has the choice of accepting, rejecting, or ignoring the waiting call. This service is defined in ANSI T1.613.

4.1.6.5.5 Call Hold. The Call Hold service allows a user to interrupt communications on an existing call and then subsequently, if desired, reestablish communications. This service is defined in ANSI T1.616.

4.1.6.5.6 Call Forwarding. The Call Forwarding service allows a served user to have the network send to another number all incoming calls for the served user's number. This service is defined in ITU-T I.252.

4.1.6.5.7 Normal Call Transfer. The Normal Call Transfer service allows a user to transfer an established call to a third party. This service is defined in ANSI T1.632.

4.1.6.5.8 Conference Call. The Conference Call service allows a user to establish calls to multiple parties, one at a time, using normal call-handling procedures. The parties may also communicate among themselves. This service is defined in ITU-T I.254, the section titled I.254.1 - Conference Calling Service Description.

4.1.6.5.9 User-to-User Signaling. The User-to-User Signaling service allows users to send and receive limited amounts of user-generated information to and from another user-network interface. This information is passed transparently (without changing contents) through the network. Users can transfer information during the establishment and clearing phases of calls. The information is transmitted in the user-user information element. The user-user information element is an optional element of the following Digital Subscriber Signaling System Number 1 (DSS1) types of messages: Alerting, Connect, Disconnect, Progress, Release, Release Complete, and Setup. This service is defined in ANSI T1.621.

4.1.6.5.10 Calling Line Identification Presentation. The Calling Line Identification Presentation (CLIP) service provides the called party with the calling line identification at call setup on all incoming calls. This service applies to both basic rate and primary rate interfaces. This service is defined in ANSI T1.625.

4.1.6.5.11 Calling Line Identification Restriction. The Calling Line Identification Restriction (CLIR) service notifies the network that the Calling Party Number is not allowed to be presented to the called party. This service is defined in ANSI T1.625. This service applies to both basic rate and primary rate interfaces.

4.1.6.5.12 Call Completion to a Busy Subscriber. The Call Completion to a Busy Subscriber service allows an authorized user, A, who encounters a busy destination, B, to be notified when the busy destination, B, becomes idle. The network

reinitiates the call to destination B if user A desires. This service is defined in ANSI Drafts T1S1.1/92-253 and T1S1.2/92-323.

4.1.6.5.13 Message Waiting Indicator Control and Notification. The Message Waiting Indicator (MWI) Control and Notification service is provided by the network to a Message Storage and Retrieval (MSR) System provider. The MSR system may request the network to provide an indication to one of its client users that messages are waiting at the MSR system. This service is defined in ANSI T1.622.

4.1.6.5.14 Line Hunting. The Line Hunting service allows a served user to enable incoming calls to a specific ISDN number to cause a search for an available hunt group member to which calls can complete. This service may be made available on demand or by subscription. This service applies to both basic rate and primary rate interfaces. This service is defined in ANSI Draft T1S1.1/92-188.

4.1.6.5.15 Explicit Call Transfer. The Explicit Call Transfer service allows a service user having two independent calls to connect together the distant parties of the two calls, thereby releasing the served user from the call, in a single user-network interaction. This service applies to both basic rate and primary rate interfaces. This service is defined in ANSI Draft T1S1/94-553.

4.1.6.5.16 Call Park. The Call Park service allows a service user to interrupt a speech or voice band data communication on an existing call, and then, re-establish communications from the same or different terminal equipment within the same Call Park Subscriber Group. A Call Park Subscribers Group is a group of Call Park subscribers designated by the service provider. The service provider may optionally group together Call Park subscribers into a Call Park Subscriber Group, in order to provide a measure of security. This service applies to the basic rate interface. Call Park is a circuit switch voice service with similar characteristics of Call Hold, except for the ability to re-establish communications from different terminal equipment. This service is defined in ANSI Draft T1S1/LB 95.05.

4.1.6.5.17 Call Deflection. The Call Deflection service permits a served user to respond to an offered call with a request to deflect the call to another number. As a subscription option, the subscriber can invoke the deflection request after answering the call. In addition, the subscriber can limit the time it takes for the deflected-to user to answer the call. If the deflected-to user does not answer within a specified time interval, the network stops the deflection attempt and returns a failure indication to the deflecting user, if the deflecting user is still associated with the call. Unlike Call Forwarding, Call Deflection allows the network to redirect a call only after

receipt of a specific user request to deflect that call. This service is defined in ANSI T1.642.

4.2 Information-transfer utility system parameters. The following ISDN system parameters listed in 4.2.1 through 4.2.5 shall apply to the information-transfer utilities portion of the DISN. These parameters are summarized here because of their impact on the design of information sources, sinks, and processors that exchange information through information-transfer utilities.

4.2.1 Information bearer channels. Base information-transfer systems shall be able to exchange multiple bearer channels over a single connection at reference point A. Below are four interface options:

a. Basic rate interface. The basic rate interface provides two 64-kbps bearer (B-) channels and one 16-kbps signaling data (D-) channel. B-channels can be used for voice or data. The D-channel is used for call control and low-speed packet data. The required method for multiplexing the 2B+D channels into a form suitable for transmission over a single twisted wire pair is provided in 5.1.1.1.1.

b. Primary rate interface. The primary rate interface provides a combination of 23B- (or 30B-) channels and one 64-kbps D-channel. The 23B+D (or 30B+D) channels will be used primarily to connect private branch exchanges (PBX) to central offices at reference point A. The primary rate interface will also be used at reference point B to interconnect local-network elements to wide-network elements. A D-channel may not be required for every primary rate interface; in this case, all 24 or 31 channels shall be available for use as B-channels.

c. High-rate channels. It shall be possible to treat multiple 64-kbps channels as a single high-rate (H) channel. Six B-channels can be treated as a single 384-kbps (H_0) channel. Twenty-three B-channels can be treated as a single 1472-kbps (H_{10}) channel. H_0 channels can be used in combination with B-channels on the same primary rate interface (PRI). Rules for time-slot assignments for high-rate signals are provided in 5.1.2.1.1.j.

d. Broadband services interface. See 5.4.

4.2.2 Timing and synchronization

4.2.2.1 Reference point A. In general, information-source bit timing shall be slaved to the local network, as described in MIL-STD-188-115, the discussion of master-slave operation. Terminal equipment connected to network elements in the base information-transfer system shall comply with ANSI T1.601, the section titled *Baud Rate, Timing, and Synchronization*. Terminal

equipment connected to network elements in the tactical information-transfer system shall comply with MIL-STD-188-115.

4.2.2.2 Reference point B. Local-network and wide-network elements that provide the reference point B interface shall provide stratum 1 clock accuracy, as defined in ANSI T1.101 and ITU-T G.811, and buffering sufficient to maintain BCI for a minimum of 24 hours. Frame synchronization, as required to demultiplex time-division-multiplexed signals, shall be provided by use of the framing bits described in ANSI T1.408 and MIL-STD-188-202.

4.2.2.3 Coordinated universal time. Systems that require time and frequency reference information based on coordinated universal time (UTC) shall comply with FED-STD-1002.

4.2.3 Tactical-to-strategic gateway functions. Gateways are a necessary solution for the interoperability of near-term tactical and strategic networks. Reference point B shall include a wide-network gateway function to achieve interoperability between switched subscribers in tactical information-transfer systems and base information-transfer systems. The gateway function requires standards for layers 1 to 3 for tactical systems using 16-kbps channels (optionally 32-kbps) and strategic systems using 64-kbps channels, as described in MIL-STD-188-105, sections 5.2 and 5.3. Since end-to-end encryption is a long-term objective of this MIL-STD, the use of gateways that require reencryption should be minimized. Gateway functions for circuit- and packet-switched connections are described in 4.2.3.1 and 4.2.3.2.

4.2.3.1 Circuit-switched voice and data. The gateway function shall include the capability to convert voice algorithms used in tactical information-transfer systems to 64-kbps pulse-code modulation (PCM) used in base information-transfer systems, in accordance with MIL-STD-188-105, section 5.5. The gateway function shall also be capable of rate adaptation, in accordance with MIL-STD-188-105, sections 5.6 and 5.9. The gateway function shall be able to accommodate additional conversion algorithms if needed in the future. The gateway function shall include the signaling-message converter, defined in MIL-STD-188-105, section 5.4, to allow tactical-switched systems to internetwork with strategic-switched systems. The signaling messages shall be examined by the gateway function. If it is determined that a secure voice or data call is being established, the gateway function shall use rate adaptation at the interface, in accordance with 4.1.6.1.4. Permanent and semipermanent circuits shall enable data transfer across the gateway at multiples of 16 kbps, up to 512 kbps, in accordance with MIL-STD-188-105, section 5.9. BCI shall be maintained.

4.2.3.2 Packet-switched data. The gateway function shall enable packet-switched data transfer across the gateway at multiples of

16 kbps, up to 512 kbps, in accordance with MIL-STD-188-105, section 5.9. BCI shall be maintained.

4.2.4 System performance. System performance standards for base information-transfer systems and wide networks shall be based on the standards for 64-kbps channels, as given in 5.7. System performance standards for tactical information-transfer systems shall be based on 5.7.

4.2.5 Network management. The Simple Network Management Protocol (SNMP), as defined in RFC 1157, shall be used until it is superseded by the Common Management Information Protocol (CMIP)/Common Management Information Service (CMIS). The objective of DISN network management is to conform to the Government Network Management Profile (GNMP) (FIPS-PUB-179), and to support the establishment, reconfiguration, and maintenance of a stable signaling and user-network environment. To achieve this objective, network management entities within each segment of the DISN shall be based on an integrated management architecture and shall employ a set of common management protocols, as defined in MIL-STD-2045-38000 and its comparison MIL-HDBK-1351. DISN network management shall provide support for the following set of common management application functions:

a. Fault management

b. Configuration management

c. Account management

d. Performance management

e. Security management

Maximum use shall be made of automated management aids to ensure effective and responsive DISN network management. Section 5.6 defines specific DISN network management requirements.

4.3 Common requirements. DISN equipment will be used in a variety of applications and environments. Acquisition specifications should contain design requirements tailored to the expected application and environment; however, the nature of military operations also dictates some degree of flexibility. Extreme care must be taken to ensure that design requirements selected from applicable DoD documents are tailored to provide the necessary flexibility. The use of commercial off-the-shelf (COTS) equipment is encouraged when the acquisition authority determines that some or all of the common requirements listed in 4.3.1 to 4.3.13 do not apply.

4.3.1 Information security. The design of information systems shall allow the incorporation of communications security (COMSEC)

and computer security (COMPUSEC) to protect information against unauthorized disclosure, transfer, modification, or destruction.

4.3.1.1 Communications security. Provisions for COMSEC shall include crypto security, transmission security (TRANSEC), emission security (EMSEC), and physical security.

4.3.1.1.1 Crypto security. Information systems shall provide internal or external crypto equipment. Digital interfaces to external crypto equipment shall be in accordance with MIL-STD-188-114.

4.3.1.1.2 Transmission security. HF radio anti-jam (AJ) systems shall comply with the TRANSEC algorithm provisions in MIL-STD-188-148. Very high frequency (VHF) radios shall comply with Joint Interoperability and Engineering Organization (JIEO) Specification 9001. Ultra high frequency (UHF) radios shall comply with Standardization Agreement (STANAG) 4372. Standards for satellite communications (SATCOM) AJ systems shall be based on existing UHF, super high frequency (SHF), and extremely high frequency (EHF) common-user DoD satellite systems.

4.3.1.1.3 Emission security. Compromising emanations shall be controlled within applicable TEMPEST criteria in the current edition of NSTISSAM TEMPEST/1-91.

NOTE: National COMSEC Engineering Memorandum (NACSEM) 5201 provides design guidance and MIL-HDBK-232 provides installation guidelines for compromising emanations.

4.3.1.1.4 Physical security. Systems shall have appropriate tamper-resistant design features and tamper-detection mechanisms.

4.3.1.2 Computer security. Computer systems shall comply with applicable provisions of DoD 5200.28-STD and the security standards identified in the TAFIM.

4.3.2 Electromagnetic compatibility. Systems and associated subsystems shall be designed to achieve intrasystem and intersystem electromagnetic compatibility (EMC). There shall be no emissions by any item of the subsystem or system beyond the tolerances established in MIL-STD-461. Techniques used to measure and determine EMC characteristics shall comply with the applicable requirements of MIL-STD-462. Equipment and subsystems should be designed in accordance with applicable EMC guidance in MIL-HDBK-235, MIL-HDBK-237, MIL-HDBK-241, and MIL-HDBK-253. The EMC program must address both emissions and susceptibilities, not just emissions. Future specific electromagnetic emission requirements will require tailoring of MIL-STD-461 requirements to ensure compatibility.

NOTE: MIL-HDBK-237 provides guidance for implementing an EMC program, and MIL-HDBK-241 provides guidance for EMC enhancement

(electromagnetic interference reduction) of equipment power supplies.

4.3.3 Electronic warfare vulnerability and electronic counter-countermeasures capabilities. Electronic warfare (EW) vulnerability analyses shall be performed on all radio subsystems, beginning with the concept formulation stage. Appropriate electronic counter-countermeasures (ECCM) capabilities shall be developed to protect these systems from the applicable EW threat.

4.3.3.1 Determining the electronic warfare technical threat. EW intelligence sources shall be used to provide an EW technical threat model during the concept formulation stage of system development. The EW technical threat model shall determine if 4.3.3.2 and 4.3.3.3 are required.

4.3.3.2 Analyzing electronic warfare vulnerability. Simulation techniques should be used to assess the effects of EW on radio links. Preliminary analyses of EW effects on candidate systems should be made to help eliminate unacceptable approaches. Subsequent analyses of emerging candidate techniques and equipment should be made at several stages of development. EW vulnerability analyses should be performed in accordance with applicable department or agency directives.

4.3.3.3 Developing electronic counter-countermeasures techniques. During each phase of system development and production, available ECCM technology should be reviewed for applicability to EW vulnerability. Where necessary, ECCM capability should be made integral to the system design. It should not be assumed that ECCM remedies can be applied at later stages of system development. EW and ECCM test requirements shall be stated in applicable system specifications. HF radio AJ systems shall comply with applicable provisions of MIL-STD-188-148. VHF radios shall comply with JIEO Specification 9001. UHF radios shall comply with STANAG 4372. Standards for SATCOM AJ systems shall be based on existing UHF, SHF, and EHF common-user DoD satellite systems.

4.3.4 Human engineering design. All information systems, subsystems, and facilities shall be designed in accordance with the applicable requirements of MIL-STD-1472 and MIL-STD-46855, and in accordance with the Human Factor Interface Style Guide (Volume 8 of the TAFIM).

4.3.5 Reliability. All systems and subsystems shall be designed to meet quantitative reliability requirements. The reliability program shall be established in accordance with the applicable requirements of MIL-STD-785. Reliability acceptance tests shall be performed in accordance with the applicable requirements of MIL-STD-781.

4.3.6 Maintainability. All equipment, subsystems, and systems shall be designed to meet quantitative maintainability requirements. The maintainability program shall be established in accordance with the applicable requirements of MIL-HDBK-470. Guidance on the performance of maintainability acceptance tests may be found in MIL-HDBK-471.

4.3.7 Survivability. Survivability is the characteristic of equipment, subsystems, and systems to withstand or avoid such damage mechanisms as blast fragments, bullets, and explosive and incendiary devices, as well as the effects of such natural phenomena as lightning, without causing a malfunction. Survivability can be enhanced by such measures as adding armor plating, duplicating and separating critical components, and simplifying the design to reduce the number of critical components. The survivability of all systems and subsystems should be assessed by performing vulnerability reduction studies in accordance with applicable department or agency directives, regulations, and instructions.

4.3.8 Climatic conditions. All equipment shall be designed to meet the applicable climatic conditions specified in MIL-STD-210. The climatic condition and induced stress requirements for an equipment or an assemblage shall be consistent with the degree of exposure anticipated for intended field applications.

4.3.9 Environmental test methods. All systems and subsystems shall be designed to comply with the applicable environmental test methods specified in MIL-STD 810.

4.3.10 Electrical measurement and test methods. Electrical measurement and test methods for communications systems shall comply with MIL-STD-188-190.

4.3.11 Grounding, bonding, and shielding. Methods and practices for grounding, bonding, and shielding of ground-based telecommunications equipment and facilities, including buildings and structures supporting tactical and long-haul communications, shall comply with the applicable requirements of MIL-STD-188-124. MIL-HDBK-419 provides practical considerations for grounding, bonding, and shielding systems.

4.3.12 Radio regulations. The use of the frequency spectrum is regulated by international agreements embodied in radio regulations published by the General Secretariat of the International Telecommunications Union (ITU), Geneva, Switzerland, and modified periodically by a World Administrative Radio Conference (WARC). These radio regulations are further qualified at the national level through such Federal Government agencies as the Interdepartment Radio Advisory Committee (IRAC), and through such military agencies as the Joint Chiefs of Staff (JCS) and the Military Communications-Electronics Board (MCEB). Military frequency planning, including joint functional frequency

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allocation tables, is established as a joint action area under the MCEB. For subsystems and equipment design, the choice and performance of the equipment, as well as frequencies and emissions of any radio subsystem, shall satisfy the provisions of those radio regulations. Therefore, radio subsystem designers and users are required to have adequate familiarity with these regulations. Final approval of frequency bands, operating modes, and equipment characteristics within DoD rests with the MCEB.

4.3.13 Radio frequency spectrum characteristics. The spectral characteristics of all radio frequency (rf) transmitters, receivers, and antennas shall be measured in accordance with the applicable requirements of MIL-STD-449.

4.3.14 Conformance testing. ISO 9646 and ITU-T X.290 shall be used for the conformance testing methodology and framework to ensure that conformance testing produces correct and consistent results. DoD Communications Protocol Standards (DCPS) conformance testing shall be in accordance with MIL-HDBK-1350-2. Acquisition agencies are cautioned that successful conformance testing of an equipment will not guarantee interoperability with all other equipment that also passed conformance testing. Conformance testing is not a substitute for interoperability testing.

4.3.15 Interoperability testing. Testing shall be performed to successfully demonstrate that systems successfully interoperate. DCPS interoperability testing shall be in accordance with MIL-HDBK-1350-2. The Joint Interoperability Test Center (JITC) is currently developing an interoperability testing guide. When this document, titled *Open Systems Environment Standards Conformance and Interoperability Testing Methodology*, is published, it will codify the processes for obtaining standards conformance, interoperability certification, or both.

4.3.16 Validation. Validation of data communications protocols (DCP) shall be conducted in accordance with MIL-HDBK-1350-1.

4.4 Subsystem design considerations

4.4.1 Terminal subsystems. Digital interfaces between terminal subsystem equipment shall comply with MIL-STD-188-114 unless other standards apply.

4.4.1.1 Tactical terminal subsystems. Tactical terminal subsystems shall comply with the applicable requirements of MIL-STD-188-216 and MIL-STD-188-260.

4.4.1.2 Long-haul digital terminal subsystems. Long-haul digital terminal subsystems shall comply with the applicable requirements of ANSI standards X3.4, X3.16, and X3.25.

4.4.1.3 Facsimile subsystems. Tactical and long-haul facsimile subsystems shall comply with the applicable requirements of MIL-STD-188-161. STANAG 5000 addresses the protocols that are necessary for facsimile subsystems to work in a noise environment.

4.4.1.4 Tactical digital information links. Message formats and related information for tactical digital information links (TADIL) A, B, and C are published in CJCSM 6231. Technical characteristics for TADILs A and C are published in the MIL-STD-188-203 series; for TADIL B, in MIL-STD-188-212.

4.4.1.4.1 TADIL A subsystems. Technical characteristics of TADIL A subsystems shall comply with applicable requirements of MIL-STD-188-203-1.

4.4.1.4.2 TADIL B subsystems. Technical characteristics of TADIL B subsystems shall comply with applicable requirements of MIL-STD-188-212.

4.4.1.4.3 TADIL C subsystems. Technical characteristics of TADIL C subsystems shall comply with applicable requirements of MIL-STD-188-203-3.

4.4.1.4.4 TADIL J subsystems. Technical characteristics of TADIL J subsystems shall comply with STANAGs 4175 and 5516.

4.4.1.4.5 ATDL-1 subsystems. Technical characteristics of Army Tactical Data Link 1 (ATDL-1) shall comply with CJCSM 6231.

4.4.1.5 Formal record traffic. Until such time that the Defense Message System replaces the legacy Formal Record Traffic system, the interoperability standards described in 4.4.1.5.1 to 4.4.1.5.4 shall apply.

4.4.1.5.1 Mode I. MIL-STD-188-171 will provide the Mode I channel coordination procedure for synchronous, simultaneous, duplex data transfer over terrestrial links.

4.4.1.5.2 Mode II. MIL-STD-188-172 will provide the Mode II non-ARQ channel coordination procedure for asynchronous, simultaneous, independent, duplex data transfer.

4.4.1.5.3 Mode V. MIL-STD-188-173 will provide the Mode V ARQ channel coordination procedure for asynchronous, simultaneous, independent, duplex data transfer.

4.4.1.5.4 Mode VI. MIL-STD-188-174 will provide the Mode VI ARQ channel coordination procedure for synchronous, simultaneous, duplex data transfer.

4.4.2 Transmission subsystems. Transmission subsystems include fiber optic cables, metallic lines, and satellite and terrestrial radios.

4.4.2.1 Long-haul transmission subsystems. Long-haul transmission subsystems shall comply with the performance requirements given in 5.7.

4.4.2.2 Tactical transmission subsystems. Tactical transmission subsystems shall comply with the applicable requirements of MIL-STD-188-202.

4.4.2.3 Fiber optic communications subsystems. Long-haul fiber optic subsystems shall comply with the applicable requirements of MIL-STD-188-111. Tactical fiber optic subsystems shall comply with the applicable requirements of MIL-STD-188-111 and JIEO Specification 9109.

4.4.2.4 Metallic line transmission subsystems. Wire and cable transmission subsystems shall comply with the applicable requirements of MIL-STD-188-112.

4.4.2.5 Frequency-division multiplex subsystems. Frequency-division multiplex subsystems shall comply with the applicable requirements of MIL-STD-188-311.

4.4.2.6 Radio subsystems operating in medium frequency and lower bands. Radio subsystems operating in the medium frequency (MF) band shall comply with the applicable requirements of MIL-STD-188-141. Radio subsystems operating in the low frequency (LF) and lower bands shall comply with the applicable requirements of MIL-STD-188-140.

4.4.2.7 High frequency radio subsystems. Radio subsystems using frequencies between 3 and 30 MHz shall comply with the applicable requirements of MIL-STD-188-141. AJ transmission systems operating in the HF band shall comply with the applicable requirements of MIL-STD-188-148 and MIL-STD-188-110. HF digital voice shall use LPC at 2.4 kbps, in accordance with MIL-STD-188-113. Automatic link establishment (ALE) of HF radio links shall be accomplished using the waveforms and procedures specified in Appendix A of MIL-STD-188-141.

4.4.2.8 Very high frequency radio subsystems. Radio subsystems using frequencies between 30 and 300 MHz shall comply with the applicable requirements of MIL-STD-188-242.

4.4.2.9 Ultra high frequency radio subsystems. Radio subsystems using frequencies between 300 and 3000 MHz shall comply with the applicable requirements of MIL-STD-188-243.

4.4.2.10 Super high frequency radio subsystems. Radio subsystems using frequencies between 3 and 30 GHz shall comply with the applicable requirements of MIL-STD-188-145.

4.4.2.11 Extremely high frequency radio subsystems. Technical characteristics of EHF radio subsystems (30-300 GHz) are under consideration.

4.4.2.12 Single-channel-radio to switched-system interfaces. See 5.1.1.3.

4.5 Functional interface requirements. This section defines the scenario, network elements, and applications supported by this MIL-STD. This scenario and these applications determine which standards, options, and parameters are incorporated in this MIL-STD.

4.5.1 Scenario. The development of new systems is driven by the availability of new technology and funding for implementation. DoD procures nondevelopmental items (NDI) to meet military requirements at reduced costs. DoD will take advantage of rapid advances in commercial computer and communications technology, as well as emerging open standards, to meet future command, control, communications, and intelligence (C3I) requirements. Military-unique features must be introduced early in the commercial [illegible] systems and the provision of intelligent networks are new trends that will affect the DISN. These changes will accompany the introduction of ISDN for base and long-haul requirements, whereas the tactical system has evolved to an all-digital system based on MIL-STD-188-256. The tactical environment is expected to further evolve toward a commercial standard, as the Services upgrade their deployable systems. In the future, broadband-ISDN (B-ISDN) services will become part of the DISN. Broadband services will provide interactive and distribution services. Interactive services include bidirectional communications with real-time information transfer, such as conferencing, between users. Distribution services include broadcast services such as television or audio programs. The starting time of distributive services may or may not be controlled by the user. Tactical assets may have limited capability for broadband services, due to spectrum limitations. In the long term, ATM transport will be extended as far into the deployed environment as practical. Further extensions of services into the deployed area will be via wireless communications.

4.5.2 Network elements. This MIL-STD identifies the standards necessary for information exchange between subscribers of common-user switched systems. Subscribers may be connected to the same network, or they may be connected to different but interconnected networks. Each network may consist of different elements, as illustrated in Figures 5 and 6. Figure 6 describes the DISN Goal

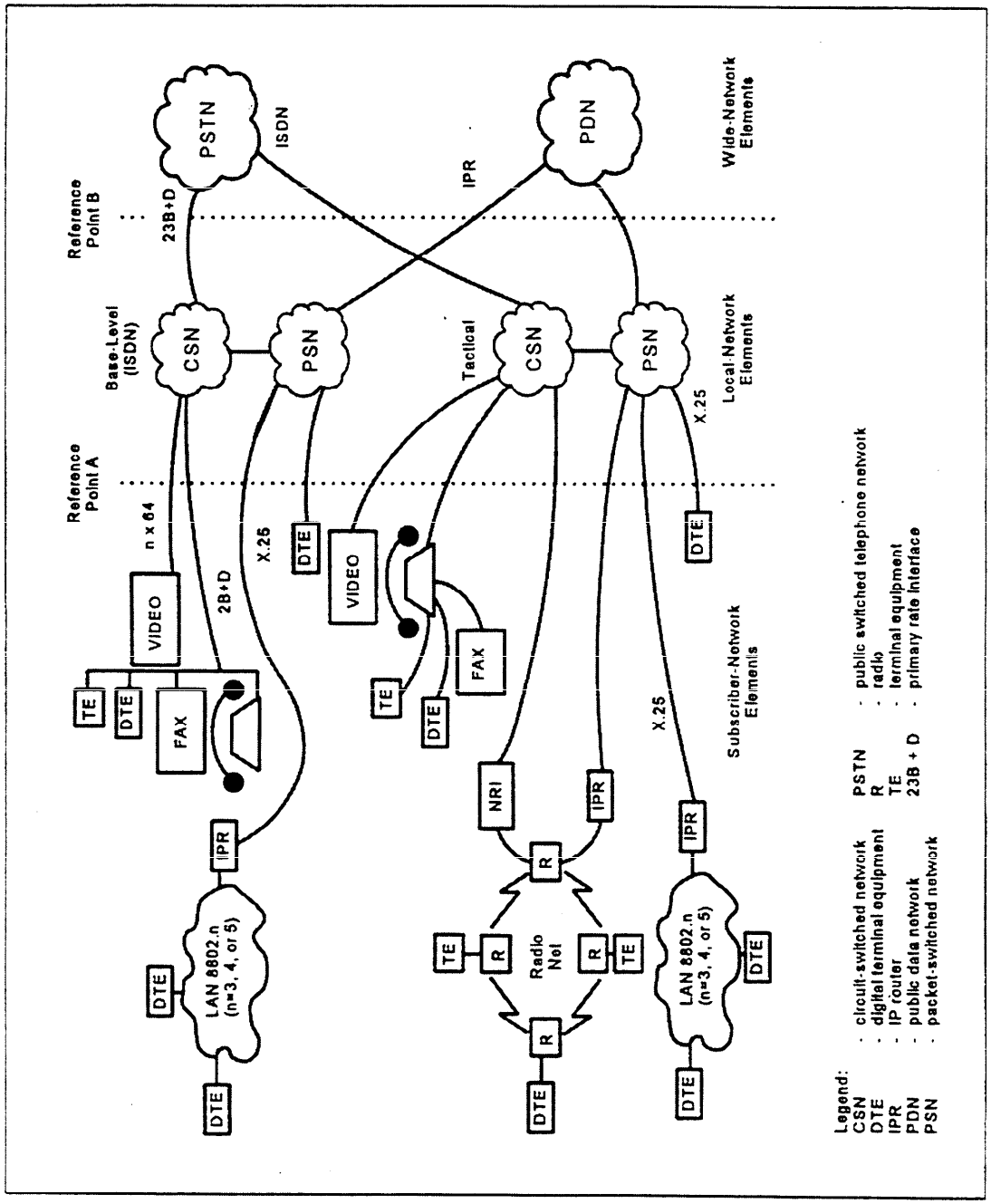


FIGURE 5. Typical DII network elements (ATM not shown).

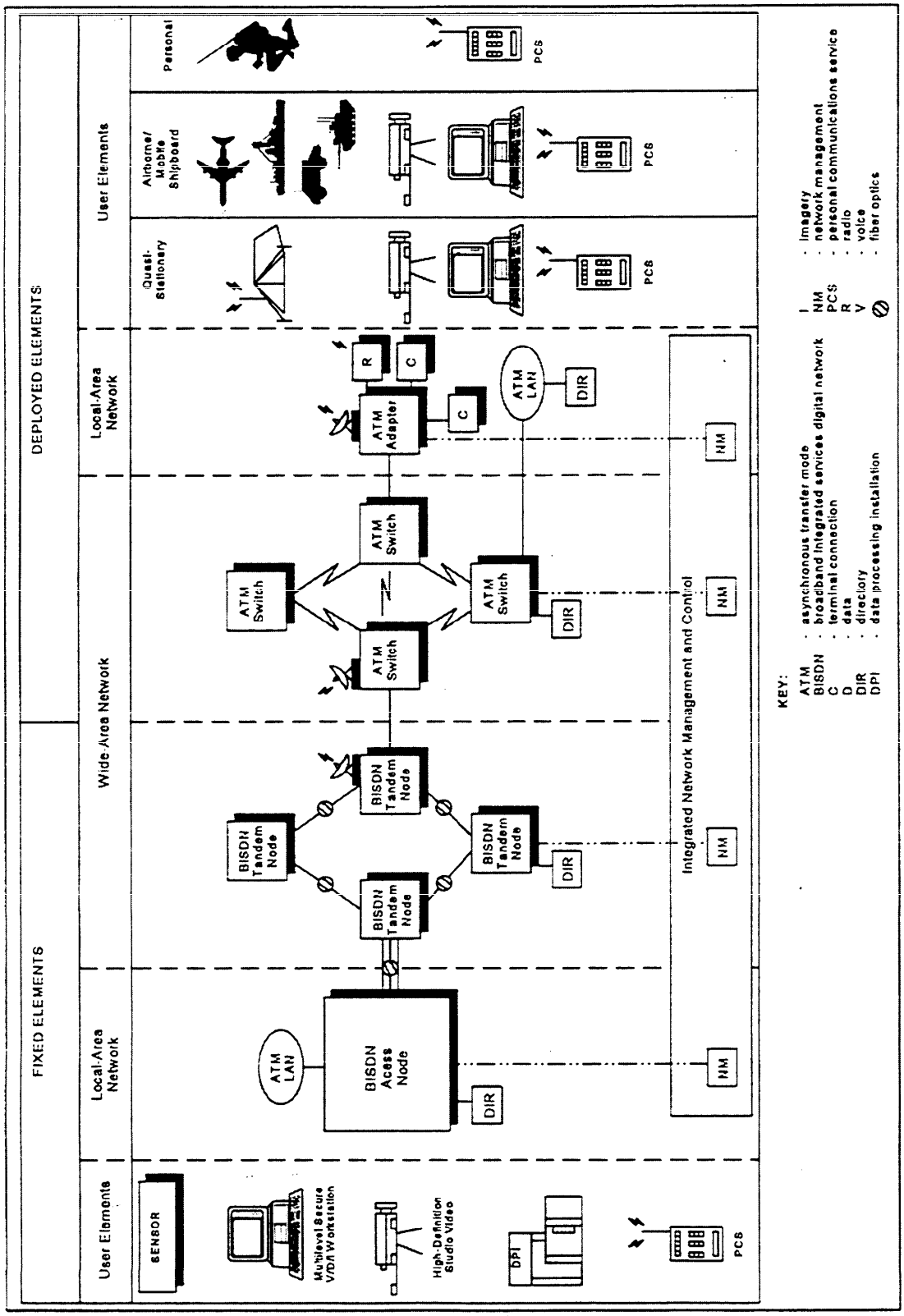


FIGURE 6. Key elements of the DISN Goal Architecture.

Architecture. These network elements are described in 4.5.2.1 to 4.5.2.4. (See the DISN Architecture.)

4.5.2.1 Subscriber-network elements. Two types of subscriber-network elements exist: subscriber terminal equipment and subscriber networks. Subscriber terminal equipment, which can be manned or unmanned, includes telephones; teleprinters; facsimile machines; data terminals (such as host computers, workstations, personal computers, digital-message entry devices, sensors, and weapons systems); video terminals; or other information sources and sinks. Subscriber networks, through a common media, provide connectivity between a limited set of subscribers. These networks can provide selective addressing, but they do not switch or route traffic. Subscriber networks may be local area networks (LAN), as defined in International Organization for Standardization (ISO) 8802-3, 8802-4, and 8802-5; or radio networks such as combat radio networks. The data communications protocols that support digital message transfer devices (DMTD) and C4I system interoperability over combat net radio are specified in MIL-STD-188-220. Subscriber terminal equipment and subscriber networks are illustrated in Figure 5. Terminal equipment (TE) designates equipment used to exchange either voice or non-voice information. DTE represents data terminals used for information-transfer applications. IPR represents routers used to transfer data between different subnetworks. *R* represents a radio terminal in a network of similar radios, for example, the Position Location Reporting System (PLRS), the Joint Tactical Information Distribution System (JTIDS), or combat net radio (CNR). For broadband systems, as illustrated by the DISN Goal Architecture of Figure 6, subscriber elements include sensors, multilevel secure voice/data/imagery workstations, high-definition studio video, data-processing installations, and personal communications services (PCS). Not shown in Figure 6 for deployed systems are the subscriber radio networks, as well as the LANs, which may be indigenous to the deployed platforms (aircraft, ships, trucks, and tanks). Subscriber terminal equipment and subscriber networks shall be directly connected to the local-network elements by cable (metallic or fiber) or by multi-role radio. The interface between subscriber-network elements and local-network elements is identified as reference point A.

4.5.2.2 Local-network elements. Local-network elements are the circuit switches, packet switches, and transmission equipment that constitute the common-user systems provided by the military Services. They provide switched service for base-level and tactical users. Local-network elements are illustrated in Figure 5. The base-level circuit-switched networks (CSN) and packet-switched networks (PSN) are based on commercial standards for ISDN. Tactical CSNs and PSNs currently interface with TRI-TAC equipment; future upgrades to tactical systems should be based on commercial standards. Compatible terminal equipment shall be able to exchange information via switched networks

comprised of local-network elements. The interface between subscriber-network elements and local-network elements shall comply with the standards described in 5.1. Tactical CSNs and PSNs within a theater of operations may be interconnected without going through the Defense Information Switched Network (DISN). As illustrated in the DISN Goal Architecture of Figure 6, local-network elements include ATM adapters, as well as broadband and digital hierarchy transmission equipment, ATM LANs, and (not shown) FDDI subnetworks. The interface between local-network elements and wide-network elements is identified as reference point B. Broadband local-network elements are based on the commercial standards identified in 5.4.

4.5.2.3 Wide-network elements. Wide-network elements are the circuit switches, packet switches, and transmission equipment provided by the Defense Communications System (DCS) and public switched telephone networks (PSTN). Wide-network elements are used to transfer information between remote local-network elements. The interface between local-network elements and wide-network elements shall comply with the standards described in 5.1.2. As illustrated by the DISN Goal Architecture of Figure 6, wide-network elements include ATM switches, broadband, and digital hierarchy transmission equipment. Broadband transmission bit-rates used in fixed networks will be based on the synchronous optical network (SONET) (see Table VIII in 5.4.1.2.1). For deployed LOS radios, bit rates will be reduced to 1.544 Mbps (DS1) to match current LOS bandwidth constraints. In the future, LOS bandwidths should include DS3. The interfaces between local-network elements and wide-network elements shall comply with the commercial standards described in 5.4.

4.5.2.4 NATO-network elements. North Atlantic Treaty Organization (NATO) network elements are the circuit switches, packet switches, and transmission equipment provided by NATO nations or those portions of the DCS that adhere to the technical standards of the host nation. Subscribers to U.S. common-user systems shall be able to exchange information with subscribers of other nations through reference point B (NATO), as illustrated in Figure 7. The interface between U.S. network elements and NATO network elements shall comply with the STANAGs and ITU-T standards described in 5.1.3. Future network elements shall comply with the Project Group (PG/6) *Tactical Communications Post-2000 Architecture*, which will lead to development of STANAGs in its final phase.

4.5.3 Military enhancements to commercial data communications protocols and standards. The intent of this MIL-STD is to adopt commercial standards for military use whenever it meets Government requirements. In some cases the commercial standards are not acceptable; thus, they require enhancements to satisfy the complete military requirement for a specific function or

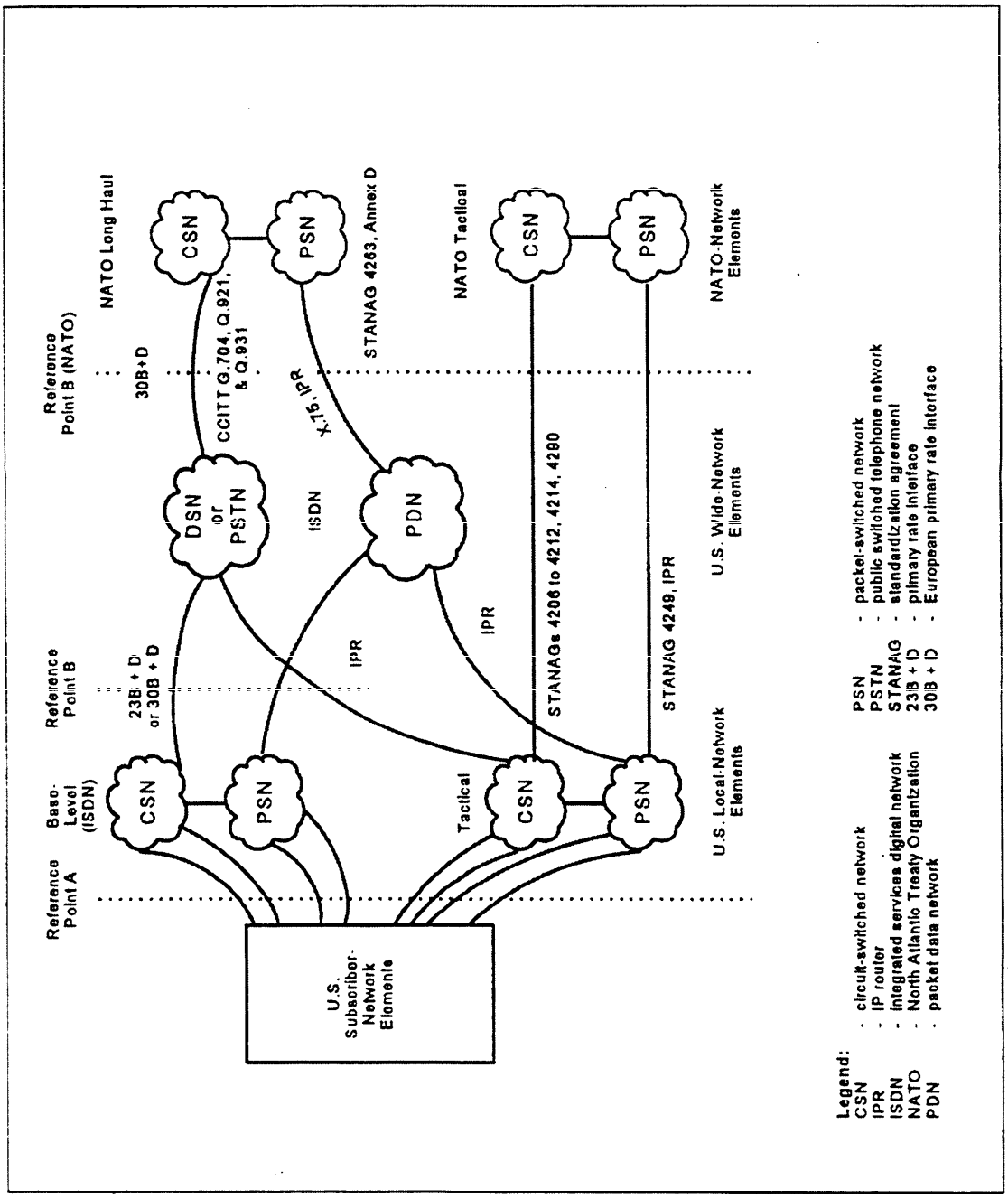


FIGURE 7. Typical DII interface with NATO network elements.

capability. These enhancements shall be found in the MIL-STD-2045 series. Eight military features have been identified in the data communications protocol area that are not adequately addressed by existing commercial data communications standards. These features are described in 4.5.3.1 through 4.5.3.8. Acquisition authorities should review the standards to ensure that these features are satisfactorily addressed and supplement their procurement documentation, as necessary.

4.5.3.1 Multihomed and mobile host systems. Multihoming is a mechanism for attaching an end system to two or more network access points so that a system setting up a call to the end system is not aware of the extra connectivity. In addition to enhancing survivability, this mechanism may be extended to "mobile hosts" such as aircraft, ships, and land vehicles during the move from one node to another.

4.5.3.2 Multi-endpoint connections (multi-addressing). To transmit data to a number of recipients (a common occurrence in signal handling), a user must establish a connection for each recipient and send a separate copy of the data across each connection. More efficient use is made of the communications resources (in particular, improved performance in terms of minimizing delay and conservation of bandwidth) if the sender has to transmit only one copy of the data. The network then takes care of routing, controlling, and distributing the data.

4.5.3.3 Internetworking. Mechanisms are required to facilitate the interconnection of various systems at the boundary point between subnetworks. Many of the interconnections would be engineered during interoperability testing.

4.5.3.4 Network and system management. Management functions are required that may be more sophisticated than those considered satisfactory for civilian networks: (a) management of broken networks, in which layers of protocols are inoperable; (b) fast responses to changes in network topology, which are essential to maintain important connections; and (c) counterattack management, to recognize and counter the effects of intelligent attack on and physical damage to the network.

4.5.3.5 Security. Protection measures are required (a) to prevent unauthorized access to the system and ensure the confidentiality of the information it carries, and (b) to preserve the integrity of the data and mitigate against denial of service.

4.5.3.6 Quality-of-service. The range of quality-of-service parameters required for military systems exceeds those currently permitted within civilian networks. The particular aim, to maximize network survivability, is to maintain an adequate quality-of-service to the users (or at least to users operating above a given priority level) in the face of a severely damaged

or partitioned network. A military requirement exists for an ultimate delivery capability, whereby important communications are sustained, even at very low data rates.

4.5.3.7 Precedence and preemption. To minimize congestion, particularly in a damaged network where resources are at a premium, it is desirable to be able to allocate resources on the basis of priority levels assigned to the messages being routed through the congested area. A facility is therefore required to associate a priority level with a message. This requirement is needed for both connection-oriented and connectionless communications.

4.5.3.8 Real-time and tactical communications. Certain applications (often tactical in nature) require communications with specified time outs, which can be in the range of milliseconds to seconds. Accurate sequencing is essential. Real time may also include high demands on sequencing accuracy.

4.5.4 Enhancements for tactical environments. DoD requires use of commercial data communications protocol standards wherever possible. Most commercial data communications standards rely on "ACK/NACK" protocols for error detection and correction. This is fine for the error environment found in commercial and strategic communications networks. However, tactical communications environments are more severe than their commercial counterparts, and reliance on ACK/NACK protocols will likely result in extremely low throughput. In some cases channel quality can be improved by operational changes such as (a) increased transmitter power; (b) better antenna setting, alignment, or both; or (c) decreased transmission rate. Beyond these link engineering activities, forward error correction (FEC) codes should be used to reduce the link bit error ratio (BER). FEC techniques applied at the physical layer will provide a higher-quality service to the higher layers and yet maintain the higher-layer protocol necessary to be interoperable with strategic networks. For tactical systems working across the NATO digital interface, FEC should comply with Class 4, as defined in STANAG 4213. The appropriate standardized FEC techniques for other tactical interfaces need further study.

4.5.5 Functional profiles. To promote open digital systems, the commercial world has developed the 7-layer Open Systems Interconnection (OSI) reference model (RM), ISO 7498, with corresponding ISO and ITU-T standards. ANSI standards address the differences between North American and European implementations. The National Institute of Standards and Technology (NIST) developed the Government Open Systems Interconnection Profile (GOSIP), Federal Information Processing Standard (FIPS) PUB 146, which specified a subset of existing standards approved for Government use. GOSIP has evolved into the Profiles for Open Systems Internetworking Technologies (POSIT), FIPS PUB 146-2, which encourages the use of voluntary

open standards, including the Internet Protocol Suite (IPS) standards disseminated by the Internet Architecture Board (IAB). (FIPS-PUB-146-1 was originally used to develop this MIL-STD.) To define the end-system interface for data communications, the 7-layer OSI RM is divided into two profiles. The top 3 layers are designated the *application profile*, and the remaining lower 4 layers are designated the *transport profile*. The application and transport profiles are described in 5.2. The *relay profile* represents the interface between two different systems and consists of the lower 3 layers with a relay function that maps one system's network layer into the other system's network layer. Relay profiles occur at Reference points A and B and are described in 5.1.1 and 5.1.2, respectively.

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5. DETAILED REQUIREMENTS

5.1 Reference point standards

5.1.1 Standards for reference point A. This section defines the standards applicable at the interface between the subscriber terminal equipment (or the subscriber's network equipment) and the local-network element (reference point A). See Figure 5.

5.1.1.1 ISDN-terminal to base information-transfer system. The terminal equipment interfacing with the base information-transfer system shall comply with the existing ANSI standards and ITU-T Recommendations cited in 5.1.1.1.1 to 5.1.1.1.3.3, and shall conform to FIPS-PUB-146 for ISDN basic rate access at the user-to-network interface. This interface applies to both circuit-switched and packet-switched service.

5.1.1.1.1 Layer 1 (the physical layer). Layer 1 provides the mechanical, electrical, functional, and procedural characteristics to activate, maintain, and deactivate a physical circuit. Layer 1 allows for the transparent transmission of bits between the terminal equipment and local-network elements. The interface between the terminal equipment and local-network elements shall comply with ANSI T1.601. This interface shall support up to 2 full-duplex, 64-kbps information bearer channels; 1 full-duplex, 16-kbps signaling channel; and 1 full-duplex, 16-kbps overhead channel over a single twisted pair of telephone wires.

5.1.1.1.1.1 Physical characteristics. The wiring polarity and connector shall comply with ANSI T1.601, the section titled *Physical Characteristics*.

5.1.1.1.1.2 Transmission method. The line code used on the twisted pair of telephone wires shall be 2B1Q (2 binary, 1 quaternary), as defined in ANSI T1.601, the section titled *Transmission Method*.

5.1.1.1.1.3 Functional characteristics. The modulation rate of the 2B1Q signal shall be 80 kilobaud. The timing signal for the subscriber's terminal equipment shall be slaved to the signal received from the local-network element. The two 64-kbps bearer channels, the 16-kbps signaling channel, and the 16-kbps overhead channel shall be multiplexed in accordance with the frame structure defined in ANSI T1.601, the section titled *Functional Characteristics*.

5.1.1.1.1.4 Electrical characteristics. The subscriber's terminal equipment shall comply with the impedance and return loss, longitudinal output voltage, longitudinal balance, jitter, and dc characteristics defined in ANSI T1.601, the section titled *Electrical Characteristics*.

5.1.1.1.2 Layer 2 (the data link layer). Layer 2 defines the procedures required to establish, maintain, and disconnect the data link between the subscriber's terminal equipment and the network.

5.1.1.1.2.1 Signaling channel (the D-channel). The link access procedure on the D-channel shall comply with ANSI T1.602. T1.602 contains the complete text of ITU-T Q.920 and Q.921, which specify the frame structure, the procedure elements, the field formats, and the link access procedures (LAP) for the D-channel (LAPD). Out-of-band signaling procedures (D-channel) shall be used to negotiate a packet-switched or circuit-switched connection for each information bearer channel.

5.1.1.1.2.2 Signaling in the bearer channel. Packet-switched calls shall be connected to the local packet handler. Remaining signaling information, including the called user address, shall be provided in the bearer channel and shall comply with the link access procedures balanced (LAPB), as defined in sections 2.2, 2.3, and 2.4 of ITU-T X.25 for basic (modulo 8) operation. Connections for circuit-switched calls shall be completed based on D-channel signaling only. At the user-to-network interface, circuit-switched calls.

5.1.1.1.3 Layer 3 (the network layer). Layer 3 protocols provide the information required to route calls through the local- and wide-network elements to the destination-terminal equipment. Three types of signaling messages shall be used to control circuit-switched and packet-switched connections: call establishment, call clearing, and miscellaneous messages. A list of the messages in each category is provided in Tables I and II.

TABLE I. Messages for circuit-switched connection control.

CALL ESTABLISHMENT	CALL CLEARING	MISCELLANEOUS
Alerting	Disconnect	Information
Call Proceeding	Release	Notify
Connect	Release Complete	Status
Connect Acknowledge	Restart	Status Inquiry
Progress	Restart Acknowledge	
Set-up		
Set-up Acknowledge		